
ClimateAR: Multi-Scale Autoregressive Generative Modeling for Climate Forecasting

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Abstract

Accurate climate forecasting provides critical support for decision-making in agriculture, energy, and disaster preparedness. Current deterministic models often fail to capture climate uncertainty, while existing generative approaches oversimplify the system by neglecting key spatiotemporal dependencies and cross-scale interactions. To address these limitations, we introduce **ClimateAR**, an AutoRegressive generative model for probabilistic Climate forecasting. The framework incorporates two novel components: (1) an aligned tokenizer that bridges and aligns heterogeneous simulation and real-world data to improve transferability across domains, and (2) a mixed-scale conditioning mechanism that captures multi-scale climate interactions for robust probabilistic forecasting. Extensive evaluations on the ERA5 reanalysis dataset show that ClimateAR achieves state-of-the-art performance, improving anomaly correlation skill by 37.56% on average compared to leading baselines.

1. Introduction

Climate forecasting provides critical guidance for a wide range of applications, including resource allocation, energy management, and agricultural planning (Meehl et al., 2021). In contrast to weather forecasting, climate prediction focuses on statistical features of anomaly fields over extended periods, which are intrinsically uncertain due to the com-

plex, nonlinear, and chaotic dynamics of the Earth system. As the forecast horizon extends beyond the weather scale (e.g., >1 month), the sources of predictability of the climate system undergo a fundamental transition: short-term, grid-resolved meteorological signals diminish in influence, giving way to slower-evolving modes of internal climate variability (Trenberth et al., 2007). A prime example is the El Niño–Southern Oscillation (ENSO), whose evolution is governed by nonlinear ocean-atmosphere interactions and modulated by stochastic atmospheric transients (Amaya et al., 2025), leading to significant irreducible uncertainty. Thus, probabilistic forecasting offers more realistic descriptions of inherent uncertainty in climate systems and thus improves the predictive skill for the possibilities of occurrence of extreme events (Price et al., 2025).

Over the past decades, Numerical Weather Prediction (NWP) models have significantly advanced climate and weather forecasting by numerically solving the governing equations of atmospheric and oceanic dynamics (Hurrell et al., 2013; Zhang et al., 2019). These physics-based systems offer strong interpretability and long-term stability, rooted in first principles. However, they face inherent trade-offs between computational cost and spatial resolution, which constrain their ability to scale to high-resolution global simulations—particularly for ensemble or real-time applications (Guo et al., 2025). Recently, data-driven approaches based on deep learning have emerged as a promising alternative (Hwang et al., 2019; Lam et al., 2023; Nguyen et al., 2023; Bi et al., 2023; Liu et al., 2025; Wu et al., 2026b;a). By learning complex spatiotemporal patterns directly from reanalysis datasets, they achieve greater computational efficiency and good accuracy in short-to medium-range forecasting. Nevertheless, most existing deep learning methods operate at the grid level and produce deterministic output, failing to capture the probabilistic nature of climate variability and underrepresenting uncertainty in long-range predictions.

To address this gap, probabilistic methods based on generative adversarial networks (GANs) (Ravuri et al., 2021; Zhang et al., 2023) and diffusion models (Price et al., 2025; Brenowitz et al., 2025) have been introduced. These meth-

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ods explicitly represent uncertainty by modeling the joint distribution of climate variables, enabling ensemble predictions. Yet, they treat the climate system as an oversimplified monolithic stochastic process, implicitly model the frequency domain rather than explicitly in physical space, potentially overlooking essential multi-scale spatiotemporal correlations (Chen et al., 2023a; Shang et al., 2024; Wu et al., 2025a; Shang et al., 2026). For example, they may fail to capture the energy cascade from large-scale climate modes (e.g., ENSO) to regional weather anomalies, a key mechanism governing seasonal predictability (Amaya et al., 2025).

In this paper, we explore the visual autoregressive (AR) model (Tian et al., 2024; Xiong et al., 2024; Li et al., 2024; Han et al., 2025) for the probabilistic modeling of climate forecasting. By representing images as sequences of discrete tokens across multiple scales, AR models capture rich semantic and spatial dependencies while avoiding oversimplified stochastic assumptions. Crucially, the multiscale inductive bias is particularly useful to model climate systems, where phenomena such as teleconnections (i.e., large-scale and statistically significant relationships between climate anomalies in geographically distant regions) and scale interactions are fundamental. In addition, the token classification-based generative process is naturally aligned with climate forecasting, where accuracy and physical consistency are prioritized over diversity.

Despite its theoretical promise, adapting the visual AR model to climate forecasting faces two major challenges: (1) **Highly heterogeneous meteorological data.** Different types of meteorological data, including global and local reanalysis data (e.g., ERA5 (Hersbach et al., 2023) and CERRA (Laffineur et al., 2023)), observation data (e.g., ground-based and satellite-based observations), and simulation data (e.g., CMIP6 (Eyring et al., 2016b)), exhibit diverse scales and characteristics. To fully utilize these data sources, a common training paradigm in climate research (Nguyen et al., 2023) involves pre-training on CMIP6 simulation data and fine-tuning on reanalysis data (e.g., ERA5). However, the distribution discrepancy between the simulated and reanalysis data hinders effective model transfer. Standard visual tokenizers in AR models treat inputs as homogeneous channels (Esser et al., 2021), lacking mechanisms to align semantic representations across domains. (2) **Complex conditional modeling.** Climate forecasts depend on accurately conditioning the generative process on the prior climatic state—a high-dimensional, multi-variable field with extremely high information density. Effectively integrating such complex conditions is challenging, particularly when capturing cross-scale interactions that govern large-scale predictability in climate dynamics, such as the teleconnections. In contrast, typical existing generative conditioning mechanisms, designed for low-dimensional

inputs like text prompts, lack the capacity to handle teleconnection and fail to effectively inject complex climatic constraints (Pang et al., 2024).

To overcome these challenges, we propose **ClimateAR**, the first AutoRegressive model designed for probabilistic Climate forecasting. ClimateAR incorporates a novel tokenization strategy and a mixed-scale conditioning mechanism to handle heterogeneous, high-dimensional climate data. The main contributions of ClimateAR are as follows:

- A principled **generative approach** to probabilistic climate forecasting. ClimateAR models the climate system as a multi-scale stochastic process, enabling the generation of ensemble forecasts that capture inherent climate uncertainty and provide a realistic description of climate processes.
- An **aligned tokenizer** that adopts vector quantization with segmented codebooks to represent high-dimensional climate variables efficiently. A shallow-separation and deep-sharing architecture aligns token semantics across simulated and real-world datasets, enhancing transferability.
- A **mixed-scale conditional control** mechanism that combines scale-specific local guidance with a hybrid-scale global prompt, capturing cross-scale interactions (e.g., the influence of large-scale oceanic anomalies on regional temperature) and leveraging high-information-density conditions effectively.
- Extensive experiments on ERA5 reanalysis data demonstrate that ClimateAR outperforms state-of-the-art AI and physical baselines, achieving an average 37.56% improvement in correlation skill. Notably, it exhibits strong performance in forecasting El Niño–Southern Oscillation (ENSO) indices, highlighting its ability to capture essential climate modes.

2. Related Work

Deep learning methods have demonstrated their effectiveness and efficiency for weather and climate forecasting (Ren et al., 2021; Chen et al., 2023c; Shi et al., 2025). Recent advances leverage sophisticated models to capture more representative meteorological signals, leading to improved accuracy. For example, Pangu (Bi et al., 2023), ClimaX (Nguyen et al., 2023), Aurora (Bodnar et al., 2025), Fuxi (Chen et al., 2023b) and FuXi-S2S (Chen et al., 2024) employ vision Transformers (Dosovitskiy et al., 2021), while WeatherGNN (Wu et al., 2024), MillGNN (Wu et al., 2025b), GraphCast (Lam et al., 2023) and Oneforecast (Gao et al., 2025) utilize GNNs. In addition, Fourcastnet (Kurth et al., 2023), Fourcastnet 3 (Bonev et al., 2025) and AIFS (Lang et al., 2026) leverage FNOs. Beyond purely data-driven

models, some deep learning approaches also attempt to incorporate physical knowledge to better capture the underlying dynamics of the climate system. For example, ClimODE (Verma et al., 2024) integrates conservation equations, SFNO (Benev et al., 2023) adopts the spherical characteristics of Earth, and WeatherGFT (Xu et al., 2024) uses PDE kernels. However, these models typically focus on capturing grid level patterns and produce deterministic forecasts, often neglecting the inherent uncertainty of the climate system. To address this gap, several approaches have explored generative models for stochasticity modeling. Notable examples include GANs-based methods (cDCGAN (Sha et al., 2024) and cGAN (Rampal et al., 2024)), and models based on diffusion processes (GraphEFM (Oskarsson et al., 2024), GenCast (Price et al., 2025), and GenAI (Lopez-Gomez et al., 2025)). However, these methods often treat the climate system as an overly simplified stochastic process, potentially overlooking critical multi-scale spatiotemporal correlations.

3. Preliminaries

Problem Definition. Following standard climate forecasting settings (Meehl et al., 2021; Arias et al., 2021), we adopt monthly-averaged meteorological data as both input and output of the model. The input is an initial climate state $\mathbf{X}_t \in \mathbb{R}^{C \times H \times W}$ at time step t , where C represents the number of meteorological variables, H and W represent the height and width of the global latitude-longitude grids, respectively. For the generative model, the forecasting task is to model the probabilistic distribution of all variables at the subsequent time step:

$$p(\mathbf{X}_{t+1} | \mathbf{X}_t) = \mathcal{F}(\mathbf{X}_t, \Theta), \quad (1)$$

where $\mathcal{F}$ represents the neural network, and Θ is the set of its learnable parameters. Then, long-term forecasts $\mathbf{X}_{t+T}$ can be obtained through T -step iterative forecasting process.

Simulation to Real-World Transfer. Existing works have demonstrated that pre-training models with simulated data can significantly enhance performance (Nguyen et al., 2023). Due to the limited availability of long-term observational and reanalysis data, it is challenging to train deep climate models from scratch with sufficient generalization ability. To overcome this data scarcity, we leverage large-scale simulated data from the Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016b) during pre-training. This allows the model to learn robust physical patterns and climate dynamics, providing a strong foundational prior. Subsequently, we transfer the model to real-world forecasting by fine-tuning on the ERA5 dataset (Hersbach et al., 2023), augmented with ocean variables from Ocean Reanalysis System 5 (ORAS5) (Zuo et al., 2019) to enhance physical consistency. This simulated-to-real transfer strategy

effectively bridges the distribution gap while capitalizing on the complementary strengths of simulated and observed climate data. Thus, our model can integrate diverse variables and achieve robust cross-domain alignment capabilities between different data sources.

Visual Autoregressive Modeling. Tian et al. (2024) proposed a novel paradigm named VAR for visual autoregressive modeling. The tokenizer first encodes the input image $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ into a continuous feature map $\mathbf{f} \in \mathbb{R}^{h \times w \times d}$, and then quantizes it into K multi-scale residual discrete token maps $(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_K)$ with their corresponding resolution $(h_1, w_1) < (h_2, w_2) < \dots < (h_K, w_K) = (h, w)$:

$$\begin{aligned} \mathbf{f} &= \varepsilon(\mathbf{I}), \quad \mathbf{res}_k = \mathbf{f} - \sum_{i=1}^{k-1} \text{up}(\mathbf{r}_i, (h, w)), \\ \mathbf{r}_k &= \arg \min_{q \in \{1, \dots, V\}} \|\mathbf{res}_k - \mathbf{c}_q\|_2, \end{aligned} \quad (2)$$

where ε is an encoder, $\mathbf{res}_k$ is the residual feature map before quantization, $\mathbf{c}_q$ represents the q -th vector in the codebook with V capacity, and $\text{up}(\cdot)$ denotes an upsampling operation. Then, the feature map $\tilde{\mathbf{f}}_k$ at the k -th scale and the reconstructed image can be expressed as:

$$\tilde{\mathbf{f}}_k = \sum_{i=1}^k \text{up}(\mathbf{r}_i, (h, w)), \quad \hat{\mathbf{I}} = \mathcal{D}(\tilde{\mathbf{f}}), \quad (3)$$

where $\tilde{\mathbf{f}} = \tilde{\mathbf{f}}_K$ is the feature map after quantization, $\mathcal{D}$ is a decoder, and $\hat{\mathbf{I}}$ is the reconstructed image. Subsequently, VAR redefines the paradigm of visual autoregressive modeling via the next-scale token map prediction:

$$p(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_K) = \prod_{k=1}^K p(\mathbf{r}_k | \mathbf{r}_{<k}), \quad (4)$$

where $\mathbf{r}_{<k}$ represents token maps prior to scale k .

4. Methodology

As illustrated in Fig. 1, ClimateAR consists of a tokenizer and an autoregressive transformer. Given an initial climate state $\mathbf{X}_t$, the tokenizer first encodes it into a sequence of multi-scale token maps $(\mathbf{r}'_1, \mathbf{r}'_2, \dots, \mathbf{r}'_K)$, which serve as conditional inputs. The autoregressive transformer then generates the forecast iteratively: starting from a learnable start token $[\mathbf{S}]$, it produces subsequent token maps scale by scale, with each step conditioned on previously generated tokens and the corresponding scale-specific features. To further guide the generation process globally, the conditional token maps are compressed into a hybrid-scale prompt that acts as a prefix across all scales. Finally, the decoder reconstructs the future climate state $\mathbf{X}_{t+1}$ from the full set of predicted token maps.

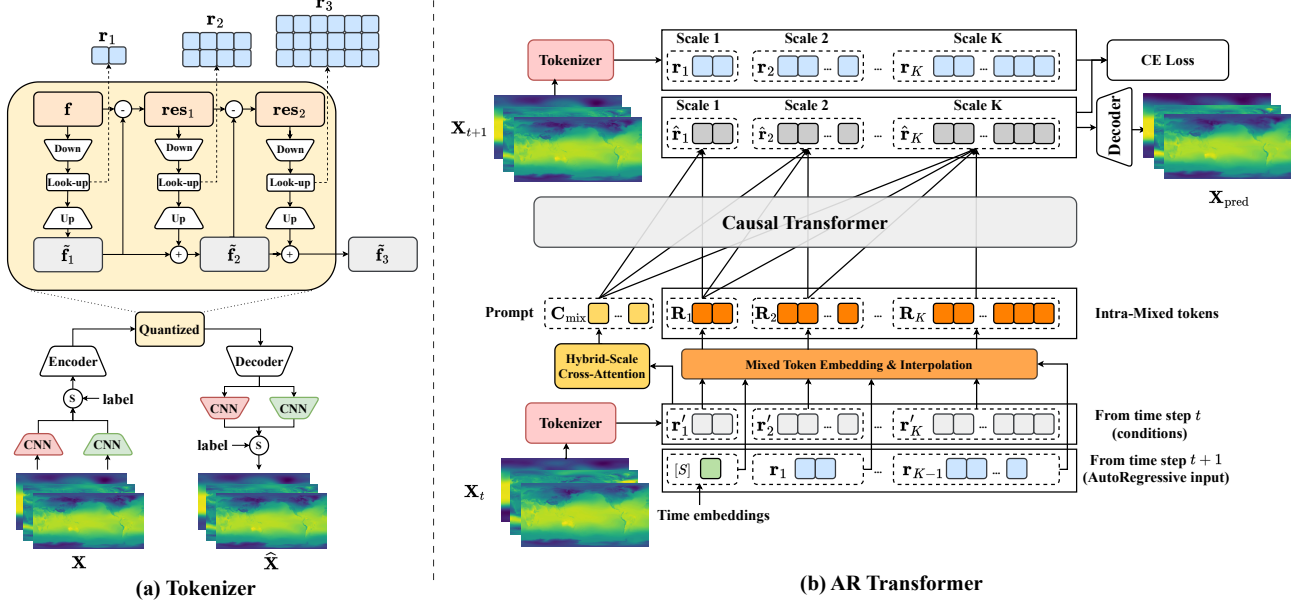


Figure 1. The architecture of ClimateAR consists of two key components: (a) a VQ multi-scale tokenizer where S replaces a selection operation by label of data source; (b) a decoder-only autoregressive transformer where CE represents the cross-entropy loss function.

4.1. Tokenize Heterogeneous Meteorological Data

High-Dimensional Feature Extraction. Following vector-quantized (VQ) methods, we encode multivariate climate data $X_t \in \mathbb{R}^{C \times H \times W}$ into a sequence of multi-scale discrete token maps ($r_1, r_2, \dots, r_K$). Given the high dimensionality and intense information of typical meteorological variables, we adopt a partitioned quantization strategy to enhance codebook capacity and reduce latent feature dimension. Specifically, the residual feature $\text{res}_k \in \mathbb{R}^{h \times w \times d}$ at scale k is split along the channel dimension into N disjoint segments, and each of them is quantized independently using a separate codebook of size V :

$$\text{res}_k = [\text{res}_k^{(1)}, \text{res}_k^{(2)}, \dots, \text{res}_k^{(N)}], \quad \text{res}_k^{(n)} \in \mathbb{R}^{h \times w \times d_n} \quad (5)$$

$$r_k^{(n)} = \arg \min_{q \in \{1, \dots, V\}} \|\text{res}_k^{(n)} - c_q^{(n)}\|_2, \quad (6)$$

where $\sum_{n=1}^N d_n = d$. This partition increases the effective codebook capacity from V to V^N while reducing the per-segment feature dimension to d/N , making the model both expressive and computationally tractable. Each token $r_k^{(i,j)}$ is thus represented by N disjoint labels during autoregressive training.

Cross-Domain Alignment. To enable knowledge transfer from simulation-based pre-training to real-world forecasting, we design an **aligned VQ tokenizer** that projects heterogeneous climatic data—from simulated and real-world sources—into a shared semantic space. As illustrated in Fig. 1, we employ domain-specific convolutional layers in

the initial stages of the encoder and the final stages of the decoder to handle low-level distribution discrepancy, while sharing deep network layers and codebooks across domains to ensure high-level climate patterns are encoded consistently. This “shallow-separation, deep-alignment” design promotes learning transferable and domain-invariant representations, improving generalization when fine-tuning on real observations.

4.2. Conditional Control

To effectively incorporate multi-scale climatic dependencies into the generative process, we condition the autoregressive generation with multi-scale information from the previous state. Specifically, the tokenizer encodes both the initial state X_t and the target state X_{t+1} into multi-scale token sequences ($r'_1, r'_2, \dots, r'_K$) and ($r_1, r_2, \dots, r_K$), respectively. The conditional distribution $p(X_{t+1} | X_t)$ can be expressed autoregressively over scales:

$$\begin{aligned} p(X_{t+1} | X_t) &= p(r_1, r_2, \dots, r_K | r'_1, r'_2, \dots, r'_K) \\ &= \prod_{k=1}^K p(r_k | r_{<k}, r'_{\leq K}). \end{aligned} \quad (7)$$

Intra-scale Mixed Token. Directly conditioning on all scales poses convergence challenges due to the high information density of climate states. We therefore introduce an *intra-scale mixed-token* mechanism that fuses the autoregressive feature $\tilde{f}_{k-1}$ (reconstructed from tokens $r_{\leq k-1}$ through Eq. (3)) with the conditional feature $\tilde{f}'_k$ from $r_{\leq k}$ at each scale. This allows the model to approximate the

conditional distribution as:

$$\mathbf{R}_k = \text{Concat}\left(\text{down}(\tilde{\mathbf{f}}_{k-1}, (w_k, h_k)), \text{down}(\tilde{\mathbf{f}}'_k, (w_k, h_k))\right),$$

$$p(\mathbf{r}_k \mid \mathbf{r}_{<k}, \mathbf{r}'_{\leq K}) \approx p(\mathbf{r}_k \mid \mathbf{R}_{\leq k}),$$

where $\text{down}(\cdot)$ performs spatial interpolation to match the target token resolution and $\mathbf{R}_k$ represents the mixed token at scale k . This approximation effectively maintains scale-wise physical consistency while leveraging the efficiency of a standard autoregressive framework.

Hybrid-Scale Prompt. While the intra-scale mixed token aligns conditional information at corresponding scales, the autoregressive process inherently lacks access to subsequent finer-scales ($\mathbf{r}'_{>k}$) conditional information. This presents a significant limitation for climate forecasting, as the evolution of large-scale climate phenomena (e.g., ENSO) is often modulated by finer-scale processes (e.g., regional convective activity or oceanic eddies). To capture these interactions, we propose a hybrid-scale prompt $\mathbf{C}_{\text{mix}}$ that compresses global, multi-scale information from the entire conditional token sequence into a global context using cross-attention:

$$\mathbf{C}_{\text{mix}} = \text{Attention}(q = \mathbf{q}, kv = (\mathbf{r}'_1, \mathbf{r}'_2, \dots, \mathbf{r}'_K)), \quad (8)$$

where $\mathbf{q}$ is a learnable hybrid-scale query sequence. The resulting representation serves as a prefix to the autoregressive sequence, enabling full cross-scale awareness. It is important to note that the compressed $\mathbf{C}_{\text{mix}}$ resides in a continuous space, whereas $(\mathbf{r}'_1, \mathbf{r}'_2, \dots, \mathbf{r}'_K)$ are discrete tokens. Therefore, even though $\mathbf{C}_{\text{mix}}$ is much shorter, we can still achieve nearly lossless compression of hybrid-scale information. The exact conditional generation thus becomes:

$$p(\mathbf{r}_k \mid \mathbf{r}_{<k}, \mathbf{r}'_{\leq K}) = p(\mathbf{r}_k \mid \mathbf{C}_{\text{mix}}, \mathbf{R}_{\leq k}). \quad (9)$$

Finally, we apply a decoder-only transformer to the entire token maps $[\mathbf{C}_{\text{mix}}, \mathbf{R}_1, \mathbf{R}_2, \dots, \mathbf{R}_k]$ to obtain next-scale token prediction $\mathbf{r}_k$.

Noise-Augmented Teacher-Forcing. Autoregressive models trained with teacher forcing are prone to *exposure bias*: during training, the model uses ground-truth tokens as input, while during inference it must use its own predictions, leading to error accumulation. To mitigate this issue, during training, we stochastically replace ground-truth tokens with random tokens while preserving the residual structure across scales:

$$\mathbf{res}_k = \mathbf{f} - \sum_{i=1}^{k-1} \text{up}(\mathbf{r}_i^{\text{noisy}}, (h, w)),$$

$$\mathbf{r}_k = \arg \min_{q \in \{1, \dots, V\}} \|\mathbf{res}_k - \mathbf{c}_q\|_2, \quad (10)$$

$$\mathbf{r}_k^{\text{noisy}} = \mathbf{r}_k \odot \mathbf{M} + \mathbf{r}_q \odot (1 - \mathbf{M}),$$

where $\mathbf{M}_{i,j} \sim B(1, p)$ is a mask for whether the element of $\mathbf{r}_k^{i,j}$ should be replaced by a random token $\mathbf{r}_q^{i,j}$, $\odot$ represents

Hadamard product, and more details are in Appendix A.1. Introducing such noise in autoregressive conditions narrows the gap between training and inference, and forces the model to rely more heavily on the conditional state $\mathbf{X}_t$ to reconstruct the target state, thus enhancing its ability to correct the accumulated errors.

4.3. Training Procedure

VQ-VAE Training: We first train the VQ-VAE on a mixture of real-world (ERA5, ORAS5) and simulated (CMIP6) climate data. Domain indicators are included to facilitate the cross-domain alignment described in Section 4.1. Once trained, the VQ-VAE weights are frozen, and the model serves as a fixed tokenizer for encoding input states and decoding predicted tokens. The training objective combines the VQ loss with perceptual terms:

$$\mathcal{L} = \|\mathbf{X} - \hat{\mathbf{X}}\|_2^2 + \sum_{k=1}^K \|\mathbf{res}_k - \mathbf{r}_k\|_2^2 + \lambda_s \mathcal{L}_s(\mathbf{X}, \hat{\mathbf{X}}),$$

where $\hat{\mathbf{X}}$ are the reconstructed data, $\mathcal{L}_s$ is defined as $\mathcal{L}_s(\mathbf{X}, \hat{\mathbf{X}}) = 1 - \text{SSIM}(\mathbf{X}, \hat{\mathbf{X}})$, denoting the structural similarity index measure to preserve large-scale patterns, and λ_s controls its weight.

AR Pretraining: We pretrain the autoregressive transformer on CMIP6 simulations to learn fundamental climate evolution patterns. At this stage, the model learns to predict the next-scale token maps conditioned on previous tokens and the hybrid-scale prompt. Since the tokenizer discretizes the climate state, we formulate the task as a classification problem, optimizing the cross-entropy loss $\mathcal{L}_{\text{CE}}$ over the codebook indices instead of using pixel-level regression.

AR Fine-tuning: Finally, we fine-tune the pre-trained autoregressive model on the ERA5 and ORAS5 reanalysis datasets to adapt it to real-world forecasting, utilizing the same cross-entropy objective. After fine-tuning, the model performs iterative forecasting by using each single-step forecast as the input for the subsequent time step, enabling seamless extension to long-term forecasts.

5. Experiments

5.1. Experimental Setup

Datasets. We pretrain ClimateAR using historical simulations from the Community Earth System Model version 2 (CESM2) (Danabasoglu et al., 2012), which contributed to the Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016a). The training data consist of nine ensemble members spanning 1850–2014, interpolated to a $1^\circ \times 1^\circ$ latitude–longitude grid. Model fine-tuning and evaluation are performed on the ERA5 atmospheric reanalysis and ORAS5 ocean reanalysis datasets (Zuo et al.,

2019). The input variables include 4 atmospheric fields at 4 vertical levels, 6 surface variables, and 2 oceanic variables at 6 vertical levels, totaling 34 channels. (See Appendix A.2 for the full variable list.) Missing oceanic values over land are filled with the global mean and masked during training. We use CMIP6 data (1850–2014) for pre-training, ERA5 and ORAS5 (1958–1999) for fine-tuning, 2000–2004 for validation, and 2005–2014 for testing.

Baselines. We evaluate ClimateAR against four leading data-driven methods: Pangu (Bi et al., 2023), GraphCast (Lam et al., 2023), Oneforecast (Gao et al., 2025), and ClimaX (Nguyen et al., 2023). To align with climate forecasting tasks and conduct a fair comparison, we retrain these models with all default hyperparameters on monthly averaged data. For comparison with conventional physics-based approaches, we include seasonal predictions from the German Meteorological Service (DWD) (Paxian et al., 2023). The DWD system uses a fully coupled climate model, initialized by nudging atmospheric and oceanic components toward reanalysis states. It produces monthly forecasts with lead times up to six months, evaluated here over the hindcast period 1993–2016. Due to the substantial computational resources required to retrain GenCast (Price et al., 2025) for interannual forecasting setting, we do not include GenCast as a baseline. Further baseline details are provided in Appendix A.2.2.

Metrics. We use the latitude-weighted Anomalous Correlation Coefficient (ACC) and Root Mean Square Error (RMSE), which are defined in Appendix A.3.1. Crucially, the ACC is employed as the **principal metric** for assessing forecasting skill. It quantifies the spatial and temporal agreement between predicted and observed anomalies—deviations from the long-term climatological mean, thus directly measuring the skill in forecasting meaningful climate variability (e.g., the development of ENSO or blocking events). By focusing on the pattern similarity of anomalies, the ACC is more robust to systematic biases in the mean state than RMSE, as minimizing RMSE tends to make the prediction close to the mean but ignore the anomalies, which is important in climate forecasting.

Implementation Details. We configure ClimateAR with a batch size of 64, 8 codebooks with a size of 4096, a hidden dimension of 1024 for both the VQ-VAE and Transformer, 16 heads and 16 layers in the Transformer architecture, and a learning rate of 0.0005 scheduled via linear warmup and cosine annealing. For the baselines, we retrained the models on monthly data with default hyperparameters to adapt to the monthly data prediction task. All these models are trained on 8 NVIDIA A800 80G GPUs. For a fair comparison, all models are pre-trained for a maximum of 50 epochs and fine-tuned for a maximum of 20 epochs with an early stop strategy. More implementation details are provided in

Appendix A.2.3 and the source code is released on GitHub¹.

5.2. Overall Performance

Baseline Comparison. We evaluate ClimateAR against baseline models on 12 key variables spanning multiple pressure levels and surface fields, including geopotential height (z500, z850, z1000), temperature (t2m, t500, t850, t1000), sea surface temperature (sst), precipitation rate (pr), sea level pressure (psl), and surface wind components (u1000, v1000). As summarized in Fig. 2 and detailed in Appendix A.4.2, ClimateAR achieves superior performance across most variables and lead times, with an average ACC improvement of 37.56% over the strongest baseline. This demonstrates the advantage of leveraging high-level, tokenized representations learned from heterogeneous data in guiding climate forecasts.

Notably, ClimateAR exhibits the most significant gains in predicting near-surface temperatures (t2m and t1000)—variables strongly influenced by large-scale climate modes such as ENSO. These results suggest that the model’s discrete token-based formulation effectively captures climate-mode dependencies, moving beyond pixel-level regression. While Pangu and ClimaX deliver competitive accuracy through ViT-based global context modeling, their deterministic nature limits their ability to represent systemic climate uncertainty. ClimateAR addresses this by explicitly modeling stochasticity through its autoregressive generative framework.

Visualization. Spatial distributions of ACC for sea surface temperature (Fig. 3) reveal that all models exhibit higher skill in tropical oceans, where climate variability is more predictable. ClimateAR consistently outperforms baselines in these regions, particularly over the tropical Pacific, indicating its enhanced capacity to capture dominant modes of interannual variability. This advantage stems from the model’s ability to represent spatio-temporal dependencies and uncertainty in a unified generative process.

Zero-shot Forecasting. To evaluate the transferability and generalization capability of ClimateAR, we conduct a zero-shot test on ERA5 data using the pre-trained model trained on the CMIP6 simulated data. We can observe in Table 1 that ClimateAR outperforms all the data-driven models. These results demonstrate its robust transferability and strong generalization performance.

5.3. ENSO Forecasting

For climate forecasting tasks, the El Niño–Southern Oscillation (ENSO) phenomenon is a widely monitored indicator. The performance for ENSO forecasting can be evaluated by

¹<https://github.com/Xiaoyuyu-1/ClimateAR>

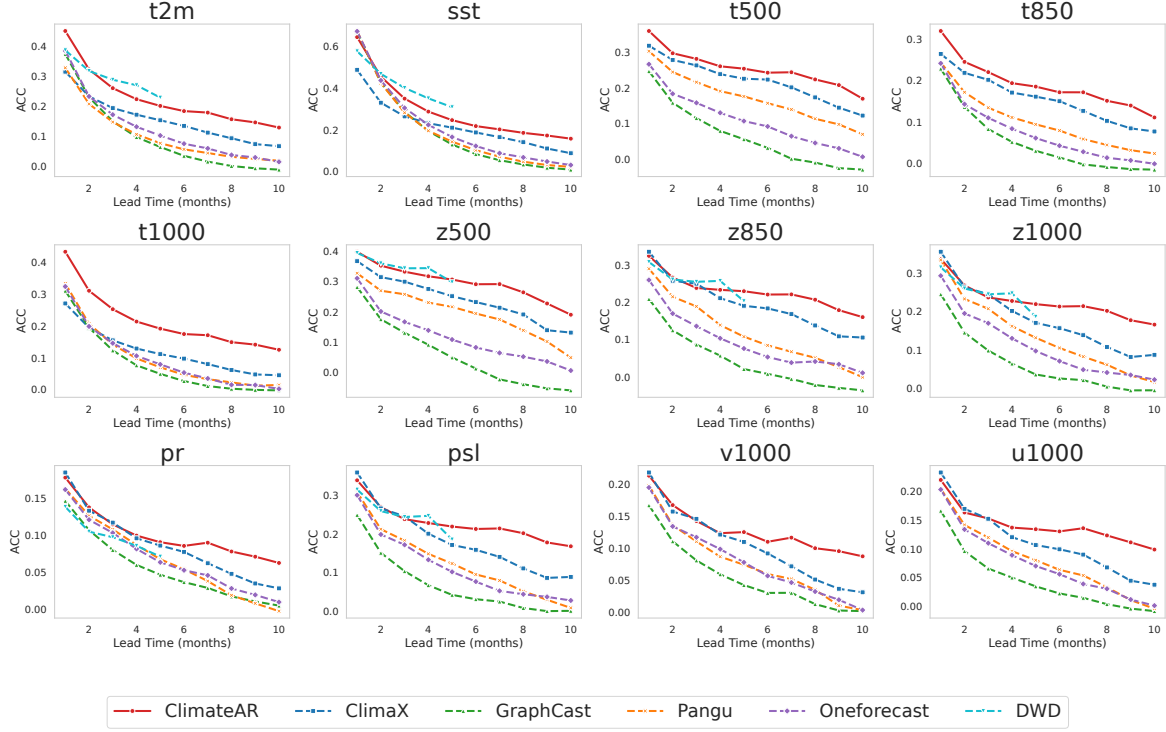


Figure 2. Comparison of ACC for ClimateAR and baselines in the global forecast from 1- to 10-month lead times.

the ENSO index calculated by the mean results of t2m in the Niño 3.4 region (170°W to 190°E and 5°S to 5°N). In Fig. 5, we compare the ACC between ClimateAR and the baselines for the ENSO index prediction task. We can observe that ClimateAR outperforms all the baselines, which demonstrates its ability to learn potential climate mode patterns.

5.4. Physical Consistency and Interpretability

To further assess the interpretability and physical consistency of the model beyond headline metrics (i.e., RMSE, ACC and CRPS), we have conducted additional analyses focusing on key large-scale climate processes.

ITCZ climatology. The intertropical convergence zone (ITCZ), a deep convective belt, migrates seasonally with atmospheric energy transport. We use the 6 mm day⁻¹ isohyet to represent the position of the ITCZ. Fig. 6, 7 shows that ClimateAR successfully reproduces its seasonal migration (proxy: 6 mm day⁻¹ isohyet) from 9°N (summer) to 2°N (winter) over the Pacific, capturing the underlying energy balance and circulation. This suggests that the model successfully represents the large-scale energy balance and associated atmospheric circulation that governs the meridional displacement of the ITCZ.

ENSO variability. ENSO is the strongest interannual vari-

ability mode in the climate system, generated by air-sea interactions in the tropical Pacific. During the El Niño peak (e.g., the 2015/16 event), ClimateAR successfully simulates the spatial patterns of both SST and precipitation anomalies in Fig. 8. Crucially, the model captures the zonal displacement of the Walker circulation, where precipitation anomalies are positioned slightly west of the SST anomalies.

These results demonstrate that ClimateAR achieves high predictive skill while maintaining strong physical consistency in energy transport and climate variability, confirming its interpretability beyond standard metrics.

5.5. Probabilistic Forecasting

Baseline Comparison. In Fig. 4, we present Continuous Ranked Probability Score (CRPS) metrics (definition in Appendix. A.3.2) of 4 key variables for global forecasting tasks. For the deterministic baselines, we obtained different ensemble forecast members through slight perturbations in the initial field. ClimateAR outperforms all baselines with an average CRPS reduction of 23.33%, demonstrating its advantage in probabilistic modeling.

Probabilistic Forecasting on ENSO. In Fig. 9, we present the performance of the probabilistic forecasting on the ENSO index. We summarize the main observations as follows: (1) twice the standard deviation of the ensemble fore-

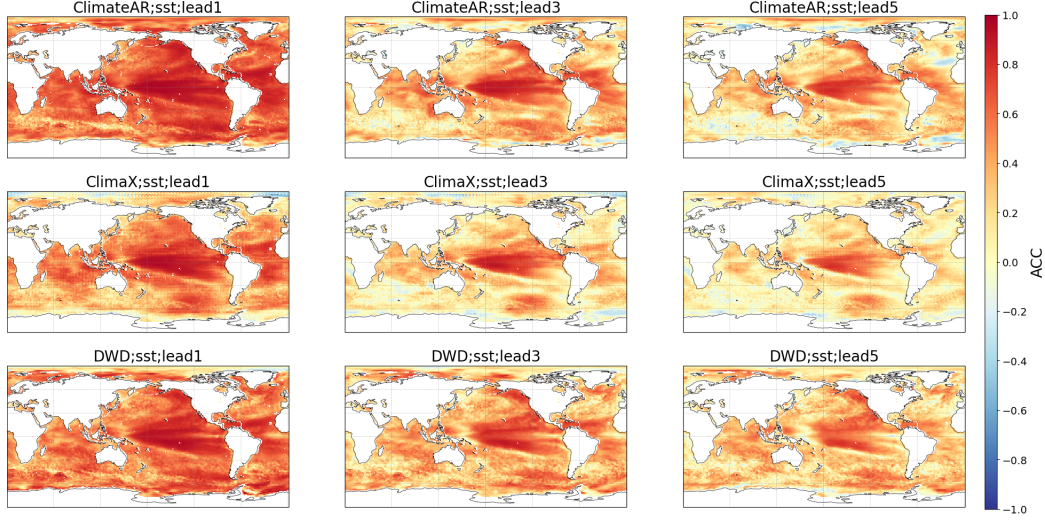


Figure 3. The global ACC distribution of sea surface temperature forecasting with 1-, 3-, and 5-month lead times in the testing set.

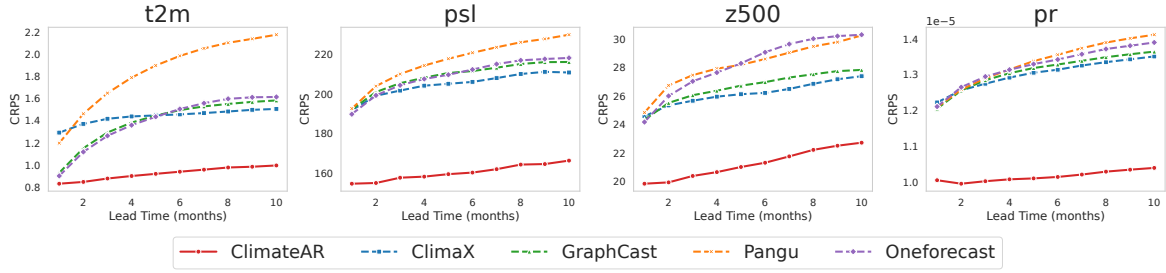


Figure 4. CRPS for ClimateAR and baselines in the global forecast from 1 to 10-month lead times. More details are in Appendix A.4.2.

Table 1. Zero-shot forecasting mean results with lead times of months 1-6 of ClimateAR and data-driven baselines. All the models are only pre-trained on simulated datasets. The best results are bolded, and the second-best results are underlined.

	RMSE ($\downarrow$)					ACC ($\uparrow$)				
	ClimateAR	GraphCast	Pangu	Oneforecast	ClimaX	ClimateAR	GraphCast	Pangu	Oneforecast	ClimaX
z500	35.333	42.493	41.263	40.663	<u>38.860</u>	0.323	0.129	0.240	0.181	<u>0.295</u>
t2m	2.011	<u>2.032</u>	2.269	2.032	2.039	0.276	0.144	0.134	0.167	<u>0.168</u>
pr ($\times 10^{-5}$)	2.037	2.041	1.986	2.053	2.003	0.112	0.068	0.091	0.088	<u>0.094</u>
psl	285.299	306.048	<u>292.689</u>	303.579	301.670	0.238	0.094	0.155	0.158	<u>0.205</u>

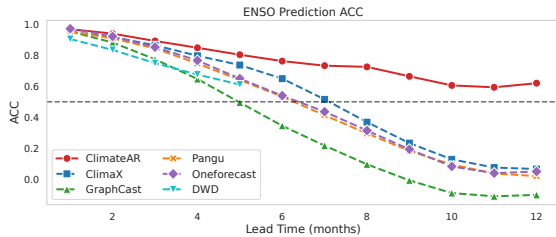


Figure 5. The ENSO index forecast results, which shows the ACC against the forecasting lead time of the ENSO index for ClimateAR and baselines.

cast can effectively capture the true observations, proving the effectiveness of ClimateAR in modeling uncertainties of the climate system; (2) starting from different time steps, within 12 months, most of the ensemble members of ClimateAR are able to effectively predict the development and dissipation of the ENSO phenomenon.

5.6. Ablation Study

Tokenizer. We consider two variants: (1) removing the cross-domain alignment module (-w/o align) and using a shared network for different data sources; (2) removing multiple codebooks and not partitioning the feature vector (-w/o part), using a single codebook for vector quantization.

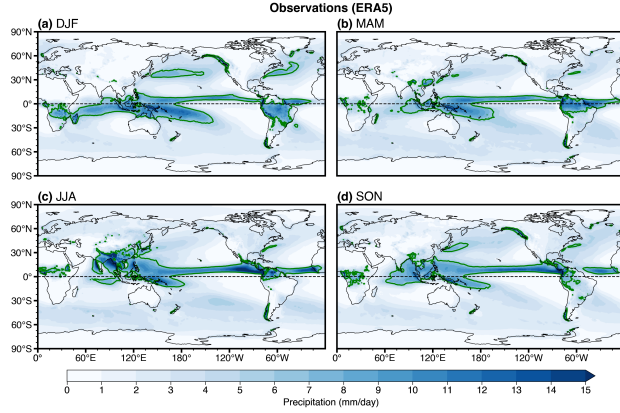


Figure 6. Seasonal cycle of climatological precipitation in observation. Green contours indicate the climatological ITCZ (mean precipitation greater than 6 mm day^{-1}). Panels show DJF (a), MAM (b), JJA (c) and SON (d).

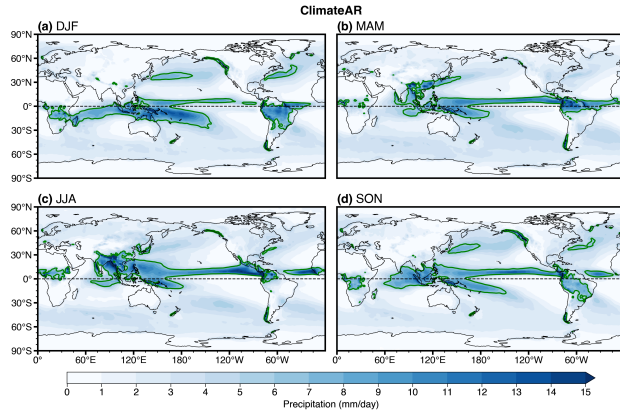


Figure 7. Same as in Fig. 6 but for the output from the ClimateAR at 1-month lead.

Conditional Control. We design three variants: (1) removing the hybrid-scale prompt (-w/o hybrid) of token maps; (2) removing the intra-scale mix tokens (-w/o intra), only using the predicted token as the input to the autoregressive function and using the conditional information as a prefix of token maps; (3) removing the noise of teach-forcing input (-w/o noise) and using correct tokens as both target and input.

Table 2. Ablation result for the 6 months averaged ACC of ClimateAR.

ACC ($\uparrow$)						
	ClimateAR	tokenizer		condition		
		w/o align	w/o part	w/o prompt	w/o intra	w/o noise
z500	0.331	0.300	0.282	0.304	0.152	0.296
t2m	0.285	0.280	0.267	0.268	0.091	0.257
pr	0.121	0.116	0.120	0.107	0.021	0.097
psl	0.258	0.236	0.241	0.218	0.061	0.213

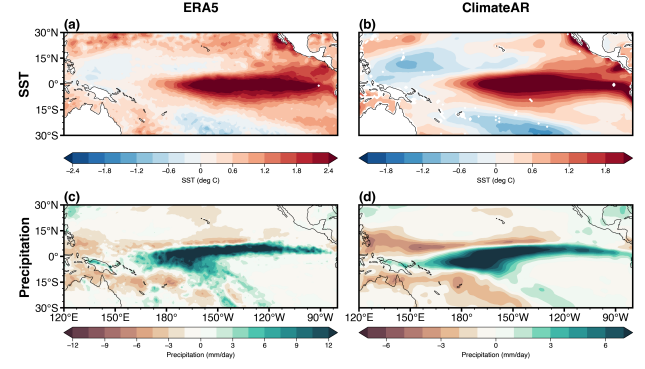


Figure 8. (a) The spatial pattern of SST anomalies ($^{\circ}\text{C}$) during the 2015/12-2016/2 derived from the ERA5 reanalysis. (b) Same as in (a) but for the output from the ClimateAR. (c) Same as in (a) but for the precipitation anomalies (mm day^{-1}). (d) Same as in (b) but for the precipitation anomalies (mm day^{-1}).

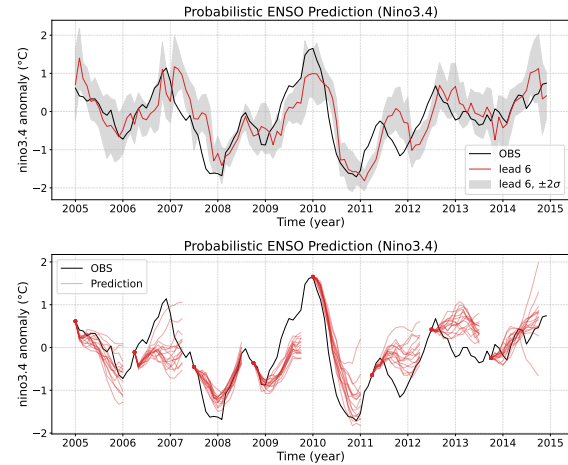


Figure 9. Ensemble forecast results of ENSO index. The first figure compares monthly average forecast results of ClimateAR and observations at 6-month lead time, where the shaded area represents the range indicating twice the ensemble standard deviation. The second figure shows the prediction over 12 months initialized at different start times, including various ensemble forecast members.

From Table 2 we can find that the lack of any key design degrades the performance of ClimateAR for the 6 months averaged ACC, verifying the effectiveness of ClimateAR.

6. Conclusions

In this work, we introduce ClimateAR, the first autoregressive probabilistic model for climate forecasting. We introduce novel tokenization and control techniques to extract and utilize high-level features of highly heterogeneous meteorological data for climate forecast generation. Extensive experimental results demonstrate that ClimateAR outperforms all competitive baselines.

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Impact Statement

This paper presents work whose goal is to advance the fields of Machine Learning and climatology. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Appendix

A.1. Additional Method Details

Tokenizer alignment and partition. For our multi-scale VQ tokenizer implementation, we incorporate two key technical components in the tokenizer pre-training procedure: partition with multi-codebook matching and labeled cross-domain data alignment. The detailed implementation algorithm is presented in Alg. 1.

Algorithm 1 Aligned tokenizer and partition.

Input: Input raw data $\mathbf{X}$, label of data source l , scale K , partitions N

Selection by label: $\mathbf{X}_l = \text{CNN}_l(\mathbf{X})$

$\mathbf{f} = \varepsilon(\mathbf{X}_l)$

Initialize: $\hat{\mathbf{f}} = 0$

for $k = 1$ to K **do**

$\text{res}_k = (\text{down}(\mathbf{f}, (h_w, h_k)))$

$[\text{res}_k^{(1)}, \text{res}_k^{(2)}, \dots, \text{res}_k^{(N)}] = \text{res}_k$

for $i = 1$ to N **do**

$\mathbf{r}_k^{(i)} = \text{look_up}(\text{res}_k^{(i)})$

end for

$\mathbf{r}_k = [\mathbf{r}_k^{(1)}, \mathbf{r}_k^{(2)}, \dots, \mathbf{r}_k^{(N)}]$

$\mathbf{f} = \mathbf{f} - \text{up}(\mathbf{r}_k, (h, w))$

$\hat{\mathbf{f}} = \hat{\mathbf{f}} + \text{up}(\mathbf{r}_k, (h, w))$

$\hat{\mathbf{f}}_k = \hat{\mathbf{f}}$

end for

$\hat{\mathbf{X}}_l = \mathcal{D}(\hat{\mathbf{f}})$

Selection by label: $\hat{\mathbf{X}} = \text{CNN}_l(\hat{\mathbf{X}}_l)$ $\hat{\mathbf{X}}$

Noise Injection. Furthermore, we introduce noise during the training phase of the autoregressive (AR) model to alleviate exposure bias inherent in autoregressive architectures. The whole process is illustrated in Alg. 2.

Algorithm 2 Tokenize data with noise.

Input: Input feature map $\mathbf{f}$, scale K , noise rate p

Initialize: target token maps $\mathbf{r}_{tgt} = []$, teach-forcing input token maps $\mathbf{r}_{in} = []$.

for $k = 1$ to K **do**

$\mathbf{r}_k = \text{Quantize}(\mathbf{f})$

$\mathbf{r}_{tgt} = \mathbf{r}_{tgt} \cup \{\mathbf{r}_k\}$

$\mathbf{r}_k^{\text{noisy}} = \text{Random_Replace}(\mathbf{r}_k, p)$

$\mathbf{r}_{tgt} = \mathbf{r}_{tgt} \cup \{\mathbf{r}_k^{\text{noisy}}\}$

$\mathbf{f} = \mathbf{f} - \text{up}(\mathbf{r}_k, (h, w))$

end for $\mathbf{r}_{tgt}, \mathbf{r}_{in}$

A.2. Additional Experimental Details

A.2.1. DATASETS

CMIP6 datasets. We use historical simulations from the Community Earth System Model version 2 (CESM2), developed by the National Center for Atmospheric Research (NCAR) (Danabasoglu et al., 2012), which participated in the Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016). The CESM2 historical simulations comprise 9 ensemble realizations with different initial conditions, covering 1850–2014, which form 9 simulated datasets used for our model pre-training. All fields were interpolated onto a $1^\circ \times 1^\circ$ grid using bilinear interpolation.

ERA5 datasets. The ERA5 dataset is the fifth-generation reanalysis product from the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 provides hourly estimates of atmospheric, land, and oceanic climate variables. The data cover the Earth on a 30-kilometer grid and resolve the atmosphere using 137 vertical levels from the surface to an

altitude of 80 kilometers. We used the period from 1958 to 2014 for model training. All fields were interpolated onto a $1^\circ \times 1^\circ$ grid using bilinear interpolation.

ORAS5 datasets. The ORAS5 dataset is the reanalysis product of the ECMWF OCEAN5 system, which is a new global eddy-permitting ocean-sea ice ensemble reanalysis system. ORAS5 provides ocean state estimates from 1979 to the present and extends historical records back to 1958, assimilating temperature and salinity in-situ observations as well as sea ice concentration data. We used the period from 1958 to 2014 for model training. All fields were interpolated onto a $1^\circ \times 1^\circ$ grid using bilinear interpolation.

Variables. Table 3 presents all the variables we use in experiments, including 4 atmospheric variables at 4 vertical levels, 6 single-level variables, and 2 oceanic variables at 6 vertical levels, totaling 34 variables. For sea surface temperature (sst), we use 5-meter sea temperature as an approximation in the corresponding tests.

Table 3. Variables used in experiments.

Type	Variable name	Abbreviation	Levels or Depth
Atmospheric	geopotential	z	200hPa,500hPa,850hPa,1000hPa
Atmospheric	u wind component	u	200hPa,500hPa,850hPa,1000hPa
Atmospheric	v wind component	v	200hPa,500hPa,850hPa,1000hPa
Atmospheric	temperature	t	200hPa,500hPa,850hPa,1000hPa
Ocean	salinity	so	5m,20m,40m,90m,200m,300m
Ocean	sea temperature	temp	5m,20m,40m,90m,200m,300m
Single	2 metre temperature	t2m	
Single	u wind stress component	tauu	
Single	v wind stress component	tauv	
Single	precipitation rate	pr	
Single	sea level pressure	psl	
Single	Sea Surface Height	zos	

A.2.2. BASELINES

Pangu. Pangu is a Vision Transformer-based data-driven model with three-dimensional deep networks equipped with Earth-specific priors. Pangu processes pressure layer variables using 3D patches and single-layer variables using 2D patches, then aggregates the two to perform prediction tasks.

GraphCast. GraphCast is a Graph Neural Network-based data-driven model. GraphCast uses multi-mesh method to construct the graph and apply a stage with an encoder, processor, and decoder to learn the complex dynamics of the system.

Oneforecast. Oneforecast is a Graph Neural Network-based data-driven model. Oneforecast constructs a multi-scale graph structure and introduces an adaptive messaging mechanism with dynamic gating units for more accurate forecasting.

ClimaX. A Vision Transformer-based data-driven foundation model for weather and climate forecasting, which tokenizes variables independently and uses an aggregation module to model the dependency between variables.

DWD. We employ seasonal climate forecasts from the German Meteorological Service. The German Meteorological Service forecasting system is based on a fully coupled climate model. Its initialization is achieved by nudging the model’s atmospheric and oceanic components toward reanalysis data. The system produces monthly forecasts with a lead time of up to six months, and its hindcast period spans 1993–2016. These datasets are available through the Copernicus Climate Change Service (C3S) hosted by the European Centre for Medium-Range Weather Forecasts (ECMWF).

A.2.3. ADDITIONAL IMPLEMENTATION DETAILS

ClimateAR. In Table 4, we present the detailed hyperparameter settings of ClimateAR. In addition, our model generates multiple forecast ensemble members through probabilistic sampling over multiple steps, and we ultimately use the average of 200 members as the final forecast result.

Table 4. The detailed hyperparameter settings of ClimateAR

Hyperparameter	Tokenizer	Hyperparameter	AR Transformer
dropout	0	dropout	0
learning rate (lr)	$5e - 4$	pre-train learning rate (lr_p)	$5e - 4$
batch size (B)	128	pre-train batch size (B_p)	64
hidden dimension (d)	1024	hidden dimension (d)	1024
codebooks dimension (c)	128	noise rate (p)	0.3
number of codebooks (N)	8	transformer heads (h)	16
size of codebooks (V)	4096	transformer layers (l)	16
number of scales (K)	10	number of scales (K)	10
loss function weight (λ_s)	1	fine-tune learning rate (lr_t)	$5e - 5$
		fine-tune batch size (B_t)	64
		prompt length (L)	256

Data-driven baselines. For Pangu, Oneforecast, and GraphCast, we utilized the source code of Oneforecast from <https://github.com/YuanGao-YG/OneForecast>, retraining the model with all default hyperparameters on monthly averaged data and conducting evaluations. For ClimaX, we adopted the publicly available source code from <https://github.com/microsoft/ClimaX>, retraining and evaluating it using the same hyperparameters with $1.40625^\circ \times 1.40625^\circ$ (patch size with 4×4).

A.3. Metrics

A.3.1. ACC AND RMSE

The detailed definitions of metric ACC and RMSE are as follows:

$$\text{ACC}(\nu) = \sum_{h,w} \frac{\alpha(h)}{HW} \frac{\sum_t \hat{\mathbf{A}}_{\nu,h,w,t} \mathbf{A}_{\nu,h,w,t}}{\sqrt{\sum_t \hat{\mathbf{A}}_{\nu,h,w,t}^2 \sum_t \mathbf{A}_{\nu,h,w,t}^2}},$$

$$\text{RMSE}(\nu) = \sum_{h,w} \frac{\alpha(h)}{HW} \sqrt{\frac{1}{T} \sum_t (\hat{\mathbf{X}}_{\nu,h,w,t} - \mathbf{X}_{\nu,h,w,t})^2},$$

where $\alpha(h) = H \cos(\lambda_h) / \sum_i \cos(\lambda_i)$ is the latitude weight factor, $\mathbf{X}_{\nu,h,w,t}$ is the ground truth for variable ν at time step t in pixel (h, w) and $\hat{\mathbf{X}}_{\nu,h,w,t}$ is its prediction. The anomaly is $\mathbf{A}_{\nu,h,w,t} = (\mathbf{X}_{\nu,h,w,t} - \mathbf{C}_{\nu,t})$, where $\mathbf{C}_{\nu,t}$ is the climatology (i.e., the mean of variable ν at the time with t in long-term historical periods).

A.3.2. CONTINUOUS RANKED PROBABILITY SCORE

The detailed definition of Continuous Ranked Probability Score (CRPS) is as follows:

$$\text{CRPS}(F, x_{\text{obs}}) = \int_{-\infty}^{\infty} (F(x) - H(x - x_{\text{obs}}))^2 dx,$$

$$H(s) = \begin{cases} 0 & s < 0 \\ 1 & s \geq 0 \end{cases}, \quad (11)$$

where F is the cumulative distribution function (CDF) of the prediction x , H is the Heaviside function and denotes the CDF of the observation x_{obs} . Then we can compute the global average CRPS using latitude-weighted averaging.

$$\text{CRPS}(\nu) = \sum_{h,w} \frac{\alpha(h)}{T HW} \sum_t \text{CRPS}(\mathbf{F}_{\nu,h,w,t}, \mathbf{X}_{\nu,h,w,t}), \quad (12)$$

where $\alpha(h) = H \cos(\lambda_h) / \sum_i \cos(\lambda_i)$ is the latitude weight factor, $\mathbf{X}_{\nu,h,w,t}$ is the ground truth for variable ν at time step t in pixel (h, w) and $\mathbf{F}_{\nu,h,w,t}$ is the CDF of its prediction.

A.3.3. SPREAD SKILL RATIO

The detailed definition of metric Spread Skill Ratio(SSR) is as follows:

$$\bar{x}_t = \frac{1}{M} \sum_{m=1}^M x_t^{(m)}, \quad s_t = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (x_t^{(m)} - \bar{x}_t)^2}, \quad \bar{s} = \frac{1}{T} \sum_{t=1}^T s_t, \quad (13)$$

$$\text{RMSE}(x^M, x_{obs}) = \sqrt{\frac{1}{T} \sum_{t=1}^T (\bar{x}_t - x_{obs,t})^2}, \quad \text{SSR}(x^M, x_{obs}) = \frac{\bar{s}}{\text{RMSE}(x^M, x_{obs})}, \quad (14)$$

where $x^{(m)}$ denotes the m -th ensemble prediction ($m = 1, \dots, M$), M is the ensemble size, and $\bar{x}$ is the ensemble mean. The ensemble spread s is defined as the square root of the ensemble variance s^2 . Then we can compute the global average SSR using latitude-weighted averaging:

$$\text{SSR}(\nu) = \sum_{h,w} \frac{\alpha(h)}{HW} \text{SSR}(\mathbf{X}_{\nu,h,w}^M, \mathbf{X}_{\nu,h,w}), \quad (15)$$

where $\alpha(h) = H \cos(\lambda_h) / \sum_i \cos(\lambda_i)$ is the latitude weight factor, $\mathbf{X}_{\nu,h,w}$ is the ground truth for variable ν at time step t in pixel (h, w) and $\mathbf{X}_{\nu,h,w}^M$ is the ensemble set of its prediction.

A.4. Additional Results

A.4.1. OVERVIEW

To further demonstrate the effectiveness, robustness and generalizability of the model, we present more extensive experimental results, including the full comparison results of RMSE, ACC, and CRPS with baselines in A.4.2, visualization of more variables in A.4.3, spread-skill analysis in A.4.4, power spectral density analysis in A.4.5, multi-scale processes interactions in A.4.6, forecasting from different months in A.4.7, decadal climate simulation in A.4.8, hyperparameter study in A.4.9 and efficiency study in A.4.10.

A.4.2. COMPARISON WITH BASELINES

Comparison on RMSE. In Fig. 10, we give the comparison of RMSE for ClimateAR and baselines in the global forecast across 1- to 10-month lead times. ClimateAR outperforms all baselines with an average RMSE reduction of 9.52%. The full results demonstrate that even though ClimateAR does not rely on pixel-wise MSE loss and only learns token-level classification in a high-level feature space, it still achieves the lowest RMSE prediction error in most cases, proving the effectiveness of modeling systematic uncertainty and spatiotemporal multi-scale dependencies in long-term forecasting tasks.

Full result. In Table 5, 6, we present all detailed metrics of the key variables for the global forecasting tasks.

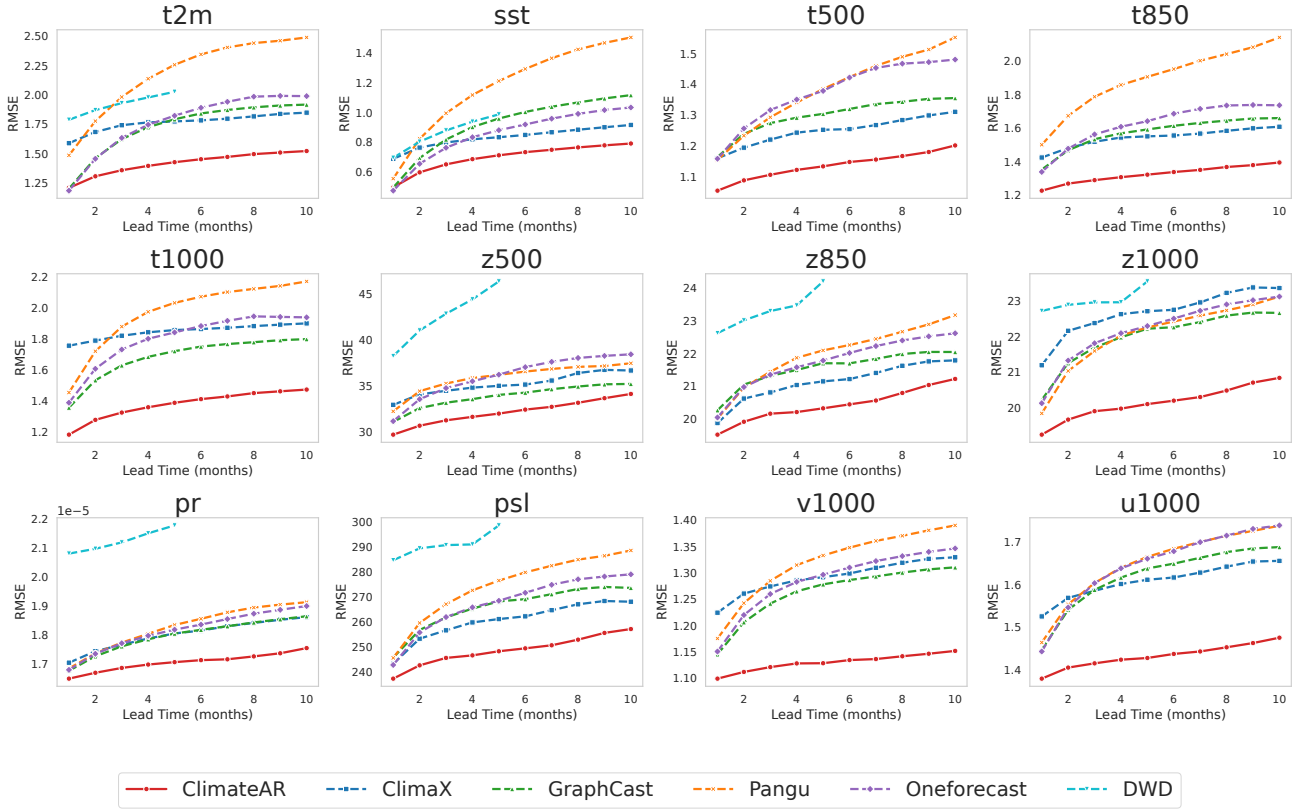


Figure 10. Comparison of RMSE for ClimateAR and baselines in the global forecast across 1- to 10-month lead times

Table 5. Global forecasting CRPS of ClimateAR and baselines. The best results are bolded.

var	lead	CRPS ($\downarrow$)					var	lead	CRPS ($\downarrow$)				
		ClimateAR	GraphCast	Oneforecast	ClimaX				ClimateAR	GraphCast	Pangu	Oneforecast	ClimaX
t2m	1	0.831	0.929	1.198	0.901	1.293	pr ($\times 10^{-5}$)	1	1.005	1.205	1.209	1.211	1.222
	2	0.847	1.152	1.463	1.118	1.371		2	0.996	1.256	1.258	1.265	1.256
	3	0.878	1.293	1.650	1.263	1.418		3	1.002	1.285	1.290	1.295	1.275
	4	0.900	1.383	1.790	1.360	1.439		4	1.008	1.304	1.314	1.314	1.292
	5	0.920	1.447	1.899	1.436	1.448		5	1.010	1.319	1.338	1.330	1.306
	6	0.939	1.496	1.986	1.506	1.457		6	1.014	1.328	1.356	1.342	1.314
	7	0.958	1.529	2.054	1.558	1.469		7	1.021	1.339	1.374	1.357	1.326
	8	0.977	1.552	2.104	1.598	1.483		8	1.029	1.349	1.390	1.372	1.335
	9	0.984	1.570	2.140	1.612	1.498		9	1.035	1.357	1.402	1.381	1.343
	10	0.996	1.583	2.177	1.615	1.507		10	1.040	1.364	1.412	1.390	1.351
	11	0.999	1.596	2.226	1.620	1.507		11	1.040	1.370	1.421	1.395	1.358
	12	1.004	1.615	2.284	1.633	1.505		12	1.043	1.374	1.432	1.401	1.364
psl	1	154.569	192.223	192.649	189.818	191.848	z500	1	19.833	24.322	24.825	24.160	24.566
	2	154.931	201.041	204.039	199.346	199.155		2	19.926	25.503	26.731	25.988	25.328
	3	157.615	205.517	210.173	204.485	201.779		3	20.377	26.042	27.453	27.019	25.655
	4	158.154	208.316	214.462	207.659	204.259		4	20.646	26.359	27.906	27.630	25.941
	5	159.407	210.686	217.911	209.790	205.218		5	21.007	26.713	28.175	28.275	26.113
	6	160.212	211.672	220.934	212.426	206.170		6	21.305	26.960	28.580	29.056	26.205
	7	161.927	213.378	223.726	215.205	208.116		7	21.748	27.283	29.026	29.632	26.484
	8	164.266	215.256	226.267	217.126	210.184		8	22.209	27.514	29.468	30.007	26.844
	9	164.532	216.191	227.955	217.776	211.274		9	22.499	27.733	29.770	30.190	27.163
	10	166.283	216.270	230.086	218.335	210.944		10	22.712	27.818	30.237	30.287	27.377
	11	165.951	216.072	232.161	219.141	211.645		11	22.756	27.840	30.860	30.343	27.565
	12	165.857	216.629	232.499	218.666	211.306		12	22.776	27.929	31.294	30.365	27.588

Table 6. Global forecasting detail results of ClimateAR and baselines. The best results are bolded, and the second best results are underlined.

var	lead	RMSE ($\downarrow$)						ACC ($\uparrow$)					
		ClimateAR	Pangu	GraphCast	Oneforecast	ClimaX	DWD	ClimateAR	Pangu	GraphCast	Oneforecast	ClimaX	DWD
t2m	1	1.210	1.487	<u>1.204</u>	1.187	1.590	1.789	0.453	0.329	0.376	0.386	0.316	0.388
	2	1.309	1.777	1.462	<u>1.457</u>	1.685	1.871	0.324	0.208	0.230	0.234	0.234	0.319
	3	1.360	1.982	<u>1.624</u>	<u>1.635</u>	1.743	1.930	<u>0.261</u>	0.148	0.150	0.172	0.195	0.289
	4	1.397	2.137	<u>1.726</u>	1.747	<u>1.767</u>	1.978	<u>0.224</u>	0.107	0.097	0.132	0.172	0.271
	5	1.428	2.256	1.793	1.823	<u>1.775</u>	2.026	<u>0.202</u>	0.077	0.064	0.103	0.154	0.230
	6	1.454	2.344	1.842	1.891	<u>1.784</u>		0.184	0.056	0.035	0.075	<u>0.135</u>	
	7	1.473	2.403	1.872	1.942	<u>1.798</u>		0.179	0.044	0.015	0.060	<u>0.112</u>	
	8	1.497	2.440	1.894	1.985	<u>1.817</u>		0.157	0.031	0.001	0.037	0.094	
	9	1.511	2.460	1.910	1.992	<u>1.839</u>		0.147	0.024	-0.007	0.029	<u>0.074</u>	
	10	1.523	2.488	1.918	1.990	<u>1.849</u>		0.129	0.018	-0.012	0.014	<u>0.067</u>	
sst	1	0.496	0.554	0.495	0.473	0.688	0.695	0.645	0.675	0.671	<u>0.673</u>	0.488	0.577
	2	0.597	0.823	0.693	<u>0.654</u>	0.763	0.803	<u>0.455</u>	0.425	0.427	0.437	0.330	0.469
	3	0.650	0.994	0.818	<u>0.762</u>	0.798	0.879	<u>0.350</u>	0.282	0.286	0.305	0.264	0.401
	4	0.686	1.117	0.902	0.833	0.819	0.938	<u>0.288</u>	0.195	0.194	0.224	0.232	0.351
	5	0.711	1.211	0.959	0.880	<u>0.833</u>	0.987	<u>0.246</u>	0.143	0.130	0.165	0.210	0.309
	6	0.733	1.292	1.003	0.919	<u>0.848</u>		0.218	0.102	0.083	0.122	0.187	
	7	0.749	1.365	1.038	0.958	<u>0.867</u>		0.202	0.071	0.054	0.088	<u>0.165</u>	
	8	0.765	1.424	1.067	0.990	<u>0.884</u>		0.185	0.047	0.034	0.068	<u>0.141</u>	
	9	0.778	1.468	1.094	1.016	<u>0.900</u>		0.173	0.031	0.018	0.048	<u>0.110</u>	
	10	0.791	1.505	1.116	1.033	<u>0.916</u>		0.158	0.022	0.009	0.031	<u>0.088</u>	
t500	1	1.055	<u>1.154</u>	1.164	1.158	1.160		0.361	0.304	0.248	0.268	<u>0.320</u>	
	2	1.088	<u>1.233</u>	1.239	1.256	<u>1.194</u>		0.298	0.245	0.159	0.184	<u>0.279</u>	
	3	1.106	1.293	1.274	1.316	<u>1.220</u>		0.283	0.217	0.115	0.159	0.264	
	4	1.122	1.341	1.291	1.351	<u>1.243</u>		0.262	0.192	0.080	0.130	<u>0.239</u>	
	5	1.133	1.385	1.304	1.378	<u>1.252</u>		0.255	0.177	0.056	0.107	<u>0.227</u>	
	6	1.148	1.423	1.320	1.422	<u>1.254</u>		0.244	0.158	0.032	0.092	0.224	
	7	1.156	1.460	1.335	1.453	<u>1.267</u>		0.245	0.140	0.001	0.065	<u>0.202</u>	
	8	1.167	1.489	1.343	1.467	<u>1.284</u>		0.225	0.114	-0.010	0.046	<u>0.174</u>	
	9	1.180	1.513	1.353	1.472	<u>1.299</u>		0.209	0.098	-0.025	0.030	<u>0.145</u>	
	10	1.201	1.552	1.355	1.480	<u>1.310</u>		0.170	0.070	-0.030	0.006	<u>0.123</u>	
t850	1	1.226	1.500	1.352	<u>1.338</u>	1.424		0.320	0.244	0.229	0.242	<u>0.264</u>	
	2	1.268	1.673	<u>1.468</u>	<u>1.477</u>	1.481		0.245	0.172	0.136	0.143	<u>0.219</u>	
	3	1.289	1.787	1.533	1.562	1.518		0.220	0.134	0.083	0.110	<u>0.202</u>	
	4	1.307	1.858	1.569	1.608	<u>1.544</u>		0.194	0.111	0.052	0.083	<u>0.171</u>	
	5	1.322	1.906	1.592	1.641	<u>1.550</u>		0.185	0.094	0.030	0.061	<u>0.161</u>	
	6	1.337	1.952	1.614	1.686	<u>1.555</u>		0.172	0.079	0.014	0.042	<u>0.150</u>	
	7	1.350	2.003	1.631	1.715	<u>1.567</u>		0.172	0.058	-0.003	0.028	<u>0.127</u>	
	8	1.367	2.042	1.645	1.735	<u>1.583</u>		0.151	0.044	-0.009	0.013	<u>0.103</u>	
	9	1.379	2.083	1.657	1.738	<u>1.599</u>		0.140	0.031	-0.014	0.007	<u>0.084</u>	
	10	1.394	2.142	1.660	1.736	<u>1.608</u>		0.111	0.023	-0.016	-0.002	<u>0.077</u>	
t1000	1	1.185	1.455	<u>1.358</u>	1.390	1.756		0.433	<u>0.333</u>	0.310	0.324	0.271	
	2	1.280	1.720	<u>1.534</u>	1.608	1.789		0.310	<u>0.207</u>	0.194	0.199	0.194	
	3	1.327	1.878	<u>1.630</u>	1.731	1.820		0.252	0.142	0.123	0.146	<u>0.155</u>	
	4	1.361	1.973	<u>1.684</u>	1.800	1.842		0.214	0.098	0.077	0.106	<u>0.129</u>	
	5	1.389	2.031	<u>1.721</u>	1.841	1.856		0.192	0.070	0.050	0.080	<u>0.112</u>	
	6	1.413	2.070	<u>1.750</u>	1.882	1.863		0.175	0.047	0.027	0.054	<u>0.098</u>	
	7	1.430	2.100	<u>1.766</u>	1.916	1.871		0.171	0.033	0.012	0.035	<u>0.080</u>	
	8	1.452	2.121	<u>1.778</u>	1.944	1.882		0.149	0.023	0.003	0.015	<u>0.062</u>	
	9	1.463	2.140	<u>1.791</u>	1.941	1.891		0.141	0.013	0.000	0.015	<u>0.048</u>	
	10	1.474	2.168	<u>1.798</u>	1.938	1.899		0.125	0.015	-0.002	0.003	<u>0.045</u>	
z500	1	29.737	32.268	<u>31.120</u>	31.201	32.974	38.268	0.398	0.327	0.282	0.311	0.367	0.394
	2	30.708	34.459	<u>32.624</u>	33.584	34.104	41.048	<u>0.352</u>	0.270	0.176	0.201	0.315	0.360
	3	31.291	35.288	<u>33.206</u>	34.789	34.498	42.861	<u>0.333</u>	0.258	0.131	0.167	0.300	0.344
	4	31.670	35.886	<u>33.592</u>	35.531	34.841	44.424	<u>0.317</u>	0.231	0.091	0.139	0.276	0.344
	5	32.022	36.229	<u>34.053</u>	36.249	35.038	46.368	0.307	0.216	0.049	0.108	0.252	<u>0.298</u>
	6	32.455	36.584	<u>34.297</u>	37.064	35.172		0.291	0.194	0.013	0.083	0.232	
	7	32.759	36.871	<u>34.674</u>	37.643	35.609		0.292	0.175	-0.023	0.064	<u>0.214</u>	
	8	33.199	37.095	<u>34.962</u>	38.062	36.429		0.264	0.139	-0.039	0.052	<u>0.191</u>	
	9	33.694	37.189	<u>35.200</u>	38.288	36.767		0.228	0.102	-0.053	0.036	<u>0.139</u>	
	10	34.156	37.485	<u>35.244</u>	38.458	36.701		0.191	0.049	-0.059	0.006	<u>0.131</u>	
z850	1	19.528	20.017	20.271	20.052	<u>19.881</u>	22.619	0.326	0.291	0.210	0.261	0.337	0.309
	2	19.920	20.961	21.053	20.978	<u>20.626</u>	23.007	0.267	0.216	0.125	0.170	0.259	0.262
	3	20.166	21.443	21.316	21.353	<u>20.814</u>	23.292	0.239	0.190	0.087	0.137	<u>0.251</u>	0.256
	4	20.215	21.867	21.497	21.579	<u>21.039</u>	23.462	0.234	0.139	0.057	0.104	0.212	0.258
	5	20.330	22.093	21.702	21.784	<u>21.150</u>	24.194	0.231	0.108	0.021	0.076	0.191	0.204
	6	20.449	22.258	21.698	22.015	<u>21.220</u>		0.222	0.085	0.008	0.054	0.184	
	7	20.567	22.442	21.835	22.225	<u>21.407</u>		0.222	0.068	-0.005	0.038	<u>0.169</u>	
	8	20.797	22.662	21.983	22.395	<u>21.625</u>		0.208	0.051	-0.022	0.041	<u>0.138</u>	

Continued on next page

Continued													
	9	21.039	22.885	22.046	22.520	21.758	0.180	0.026	-0.029	0.034	0.109		
	10	21.224	23.168	22.044	22.614	<u>21.789</u>	0.161	-0.001	-0.036	0.011	<u>0.106</u>		
z1000	1	19.243	19.839	20.222	20.125	21.193	22.719	0.336	0.334	0.245	0.293	0.355	0.314
	2	19.665	<u>21.033</u>	21.251	21.326	22.161	22.891	0.267	0.232	0.145	0.195	<u>0.263</u>	0.260
	3	19.902	21.594	21.692	21.811	22.378	22.960	0.236	0.207	0.099	0.169	0.245	<u>0.244</u>
	4	19.974	22.029	21.979	22.094	22.629	22.963	<u>0.227</u>	0.162	0.065	0.130	0.201	0.247
	5	20.101	22.256	<u>22.220</u>	22.296	22.711	23.549	0.219	0.132	0.036	0.097	0.170	<u>0.186</u>
	6	20.198	22.422	<u>22.262</u>	22.505	22.751		0.213	0.105	0.026	0.071	<u>0.157</u>	
	7	20.299	22.589	<u>22.407</u>	22.725	22.959		0.214	0.083	0.022	0.049	<u>0.138</u>	
	8	20.483	22.734	<u>22.588</u>	22.903	23.227		0.202	0.062	0.005	0.041	<u>0.108</u>	
	9	20.705	22.899	<u>22.674</u>	23.021	23.379		0.177	0.034	-0.005	0.035	<u>0.082</u>	
	10	20.838	23.123	<u>22.664</u>	23.123	23.362		0.166	0.017	-0.005	0.023	<u>0.087</u>	
pr($\times 10^{-5}$)	1	1.647	1.682	<u>1.675</u>	1.678	1.702	2.080	0.178	0.163	0.146	0.161	0.185	0.138
	2	1.668	1.735	<u>1.725</u>	1.733	1.742	2.097	0.137	0.126	0.106	0.121	<u>0.133</u>	0.104
	3	1.684	1.772	<u>1.759</u>	1.770	1.765	2.119	0.114	0.107	0.080	0.103	0.117	0.097
	4	1.696	1.803	<u>1.784</u>	1.796	1.786	2.151	0.099	0.087	0.059	0.081	<u>0.095</u>	0.085
	5	1.704	1.833	<u>1.803</u>	1.817	1.804	2.178	0.090	0.067	0.046	0.063	<u>0.086</u>	0.071
	6	1.711	1.855	1.816	1.834	1.814		0.085	0.054	0.036	0.052	<u>0.077</u>	
	7	1.714	1.877	<u>1.829</u>	1.854	<u>1.829</u>		0.090	0.038	0.028	0.045	<u>0.062</u>	
	8	1.724	1.893	1.842	1.872	<u>1.840</u>		0.078	0.018	0.017	0.028	<u>0.047</u>	
	9	1.735	1.904	1.853	1.886	<u>1.851</u>		0.071	0.007	0.010	0.019	<u>0.034</u>	
	10	1.753	1.912	1.863	1.899	<u>1.860</u>		0.062	-0.003	0.004	0.009	<u>0.028</u>	
psl	1	237.383	245.675	245.801	<u>242.856</u>	243.295	284.758	0.339	0.309	0.249	0.301	0.360	0.315
	2	242.705	259.654	256.854	255.864	253.370	289.515	0.269	0.212	0.151	0.199	<u>0.266</u>	0.259
	3	245.669	267.181	261.879	262.011	<u>256.714</u>	290.805	0.238	0.182	0.103	0.172	0.246	0.244
	4	246.687	272.706	265.405	265.906	<u>259.874</u>	291.080	<u>0.228</u>	0.148	0.067	0.133	0.201	0.247
	5	248.335	276.614	268.287	268.545	<u>261.215</u>	298.682	0.219	0.123	0.041	0.101	0.171	<u>0.185</u>
	6	249.501	279.888	269.181	271.727	<u>262.304</u>		0.213	0.095	0.030	0.076	0.159	
	7	250.728	282.558	271.174	274.927	<u>264.772</u>		0.214	0.079	0.024	0.052	<u>0.140</u>	
	8	252.975	284.993	273.249	277.126	<u>267.130</u>		0.202	0.051	0.007	0.043	<u>0.110</u>	
	9	255.679	286.517	274.057	278.237	<u>268.441</u>		0.178	0.029	-0.001	0.036	<u>0.086</u>	
	10	257.225	288.660	273.684	279.112	<u>268.147</u>		0.168	0.008	0.000	0.027	<u>0.088</u>	
v1000	1	1.099	1.175	<u>1.145</u>	1.150	1.224		0.213	0.197	0.166	0.194	0.218	
	2	1.112	1.242	<u>1.206</u>	1.220	1.260		0.167	0.136	0.110	0.134	<u>0.157</u>	
	3	1.121	1.285	<u>1.241</u>	1.260	1.274		<u>0.142</u>	0.110	0.081	0.117	0.145	
	4	1.128	1.314	<u>1.265</u>	1.283	1.285		0.123	0.086	0.059	0.098	<u>0.121</u>	
	5	1.128	1.333	<u>1.278</u>	1.296	1.292		0.125	0.074	0.042	0.078	<u>0.109</u>	
	6	1.134	1.348	<u>1.286</u>	1.310	1.299		0.110	0.059	0.030	0.057	<u>0.091</u>	
	7	1.136	1.360	<u>1.293</u>	1.322	1.309		0.116	0.052	0.030	0.046	<u>0.071</u>	
	8	1.141	1.370	<u>1.300</u>	1.331	1.319		0.100	0.035	0.013	0.032	<u>0.051</u>	
	9	1.146	1.380	<u>1.306</u>	1.340	1.326		0.095	0.010	0.003	0.019	<u>0.036</u>	
	10	1.152	1.390	<u>1.310</u>	1.346	1.329		0.087	0.003	0.002	0.003	<u>0.031</u>	
u1000	1	1.379	1.464	1.447	<u>1.443</u>	1.525		0.220	0.205	0.166	0.203	0.233	
	2	1.405	1.554	<u>1.540</u>	1.546	1.569		<u>0.163</u>	0.142	0.096	0.134	0.170	
	3	1.415	1.604	1.588	1.603	<u>1.585</u>		0.153	0.120	0.066	0.110	<u>0.152</u>	
	4	1.423	1.640	1.616	1.638	<u>1.601</u>		0.137	0.096	0.050	0.089	<u>0.120</u>	
	5	1.427	1.665	1.638	1.661	<u>1.611</u>		0.134	0.080	0.035	0.070	<u>0.107</u>	
	6	1.437	1.684	1.649	1.678	<u>1.617</u>		0.131	0.064	0.022	0.056	<u>0.099</u>	
	7	1.443	1.700	1.663	1.700	<u>1.628</u>		0.136	0.054	0.015	0.039	<u>0.090</u>	
	8	1.453	1.715	1.676	1.715	<u>1.642</u>		0.123	0.033	0.004	0.031	<u>0.068</u>	
	9	1.463	1.726	1.685	1.731	<u>1.654</u>		0.111	0.011	-0.004	0.012	<u>0.045</u>	
	10	1.475	1.738	1.688	1.739	<u>1.656</u>		0.099	-0.004	-0.008	0.001	<u>0.038</u>	

Comparison with probabilistic model. To establish a strong probabilistic baseline, we extend the deterministic backbone into an ensemble forecasting model, based on the AIFS-CRPS method. Specifically, Gaussian noise is encoded into a noise embedding, which is used as the conditioning signal for Conditional Layer Normalization, thereby injecting uncertainty into the latent space of the model. For each initial condition, the model generates an ensemble of N forecasts. The model is trained end-to-end by directly optimizing the continuous ranked probability score (CRPS) over the ensemble predictions.

Region Forecasting. In Fig. 11, we give the comparison of ACC for ClimateAR and baselines in the low latitude regional forecast (i.e., 30°S to 30°N.) across 1- to 10-month lead times. The full results demonstrate that ClimateAR can significantly extend its effective forecast step over yearly in regions with greater predictability (i.e., low-latitude region).

A.4.3. VISUALIZATION

To further demonstrate ClimateAR’s forecasting capability across different regions, we visualize the RMSE distribution of Sea Surface Temperature (sst) and the ACC distribution of other variables in Fig. 12, 13, 15, 14. From the extensive results, we observe that ClimateAR performs particularly well in regions with high predictability (e.g., the tropics and the eastern Pacific—on seasonal scales). This highlights the model’s ability to capture climate modes critical for long-term climate

Table 7. CRPS comparison of ClimateAR and AIFS-CRPS with different lead times. The best results are bolded.

Variable	Method	1	2	3	4	5	6	7	8	9	10
t2m	ClimateAR	0.831	0.847	0.878	0.900	0.920	0.939	0.958	0.977	0.984	0.996
	AIFS-CRPS	1.229	1.295	1.332	1.373	1.391	1.383	1.378	1.387	1.393	1.413
pr($\times 10^{-5}$)	ClimateAR	1.005	0.996	1.002	1.008	1.010	1.014	1.021	1.029	1.035	1.040
	AIFS-CRPS	1.077	1.104	1.124	1.141	1.152	1.159	1.165	1.169	1.173	1.176
psl	ClimateAR	154.569	154.931	157.615	158.154	159.407	160.212	161.927	164.266	164.532	166.283
	AIFS-CRPS	163.161	170.280	172.453	176.757	177.775	179.007	180.774	182.642	184.100	185.077
z500	ClimateAR	19.833	19.926	20.377	20.646	21.007	21.305	21.748	22.209	22.499	22.712
	AIFS-CRPS	21.701	22.765	23.264	23.960	24.132	24.068	24.154	24.438	24.788	25.290

prediction.

A.4.4. SPREAD-SKILL ANALYSIS

As shown in Fig. 16, both the ensemble spread and the RMSE of the ensemble increase with the increase in lead times, indicating that the uncertainty of the forecasting increases and the accuracy decreases over longer prediction time. Meanwhile, the Spread-Skill Ratio (SSR) also exhibits an increasing trend with horizon. This indicates that the growth rate of ensemble spread is faster than the growth rate of the RMSE, meaning that prediction becomes more conservative over longer prediction time. This suggests that the model’s output spread tends to increase when the forecasting uncertainty rises.

A.4.5. POWER SPECTRAL DENSITY ANALYSIS

We consider the power spectral density (Hickman et al., 2025) of the Niño3.4 index and Indian Ocean Dipole (IOD) index in Fig. 17. We can find that ClimateAR successfully captures the primary peaks of different climate indices at interannual timescale.

A.4.6. MULTI-SCALE PROCESSES INTERACTIONS

Global and Regional Scale. To evaluate the ability of ClimateAR to capture interactions across multiple scales, we examine the relationship between ENSO and global mean surface temperature (GMST) anomaly. In Fig. 18, we present the time series of the Niño3.4 index and GMST anomaly from both observations and predictions, together with their corresponding regression coefficients. The regression coefficients between ENSO and GMST in the predictions is nearly identical to those in the ERA5 dataset (difference less than $0.05K$), and the predicted values accurately reproduce the amplitude and phase of the observed variations. This shows that the model effectively captures the short-term temperature fluctuations associated with ENSO events and successfully reproduces the expected lagged response of GMST to ENSO forcing (typically about 3-6 months).

Regional and Local Scale. Fig. 19 shows the spatial distribution of regression coefficients between ENSO and pr/t2m anomaly at each grid. The predicted patterns closely resemble those observed. The model accurately identifies the characteristic teleconnections associated with ENSO and captures the positive or negative grid-point-level responses to ENSO intensity across the global domain.

Overall, GMST represents large-scale global variability, ENSO indices represent mid-scale variability, and grid-wise precipitation/t2m anomalies represent small-scale regional fluctuations. The high consistency between predicted and observed relationships across all these scales indicates that ClimateAR effectively models the interactions among climate processes and successfully captures cross-scale variability.

A.4.7. FORECASTING FROM DIFFERENT MONTHS

In Fig. 20, we use each of the 12 months in a year as a forecasting start point with lead times of 1, 6, and 10 months, respectively, and compute the average t2m forecast ACC values. It can be seen that ClimateAR achieved the best ACC level in most months. For some months that are difficult to predict (e.g., the months when the ENSO starting phase is reset), we can also get high ACC performance.

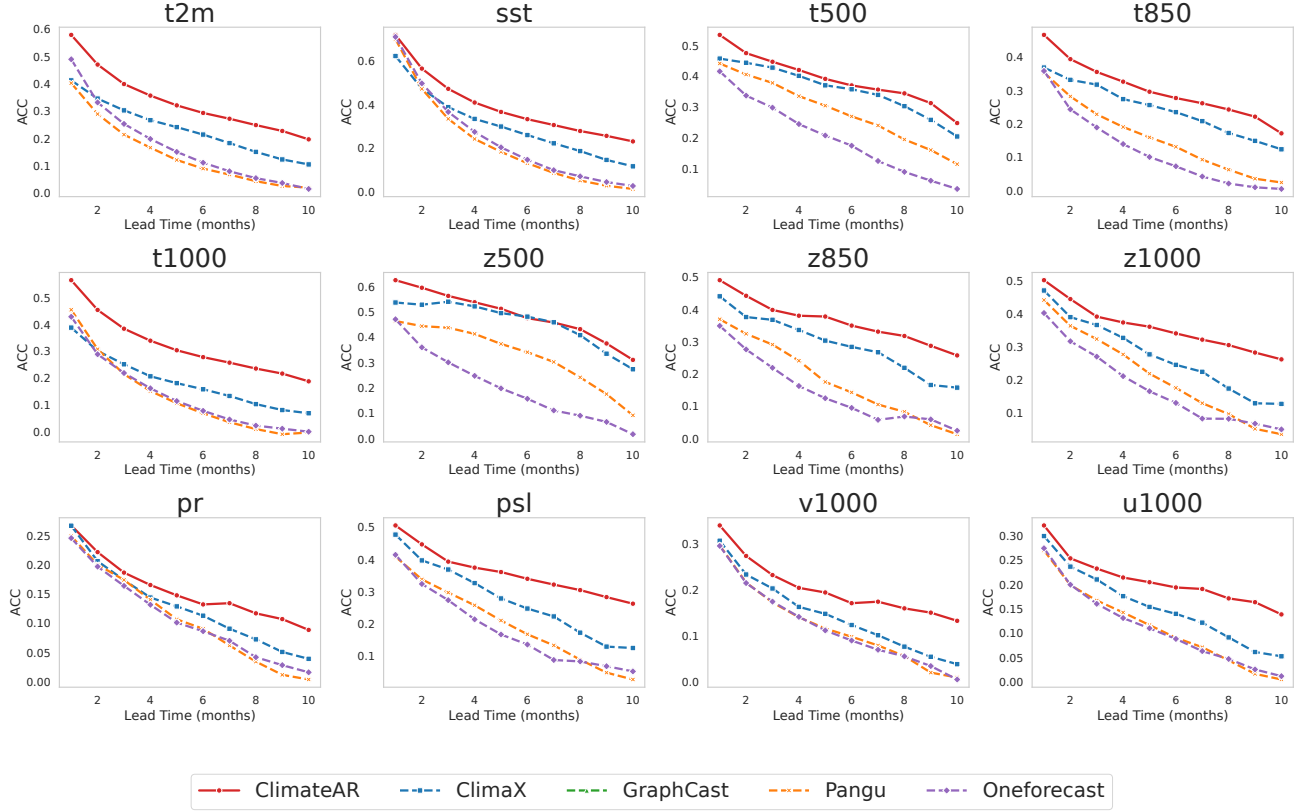


Figure 11. Comparison of ACC for ClimateAR and baselines in the low latitude regional forecast across 1- to 10-month lead times.

A.4.8. DECADEAL CLIMATE SIMULATION

To explore the long-term climate simulation potential of ClimateAR, we conduct a decadal forecasting experiment. We initialize the model with the January 1958 climate state and iteratively generate monthly predictions through December 2020, using SST as the only external forcing input. We evaluate the resulting multi-decadal time series by examining the annual and monthly global-mean t2m over 1958–2020 in Fig. 21. ClimateAR produces a stable and physically consistent evolution of global-mean temperature that closely follows the observed long-term trend. This shows that ClimateAR is capable of maintaining coherent large-scale climate statistics over extended periods, highlighting its effectiveness for decadal climate forecasting.

A.4.9. HYPERPARAMETER STUDY

We conducted a search for two critical hyperparameters during the model’s pre-training phase, including noise ratio p and number of partitions N . Specifically, p controls the noise ratio of the teach-forcing input during the training procedure, N controls the number of partitions codebooks and V controls the size of codebooks. To ensure fairness, other hyperparameters are fixed when a specific one is tuned. The details of different ACCs for different values of p and N are summarized in Table 8.

A.4.10. EFFICIENCY STUDY

Training Cost. The training time complexities of ClimateAR are $O(h^2w^2)$, where (h, w) is the size of feature map $\mathbf{f}$. which is consistent with traditional ViTs. To conduct a fair comparison, we uniformly set the batch size to 8 when measuring the GPU memory usage.

Inference Cost. Unlike traditional autoregressive models, the inference time complexities of ClimateAR are consistent with the complexity of the training $O(h^2w^2)$, which has been demonstrated in previous work (Tian et al., 2024). When the

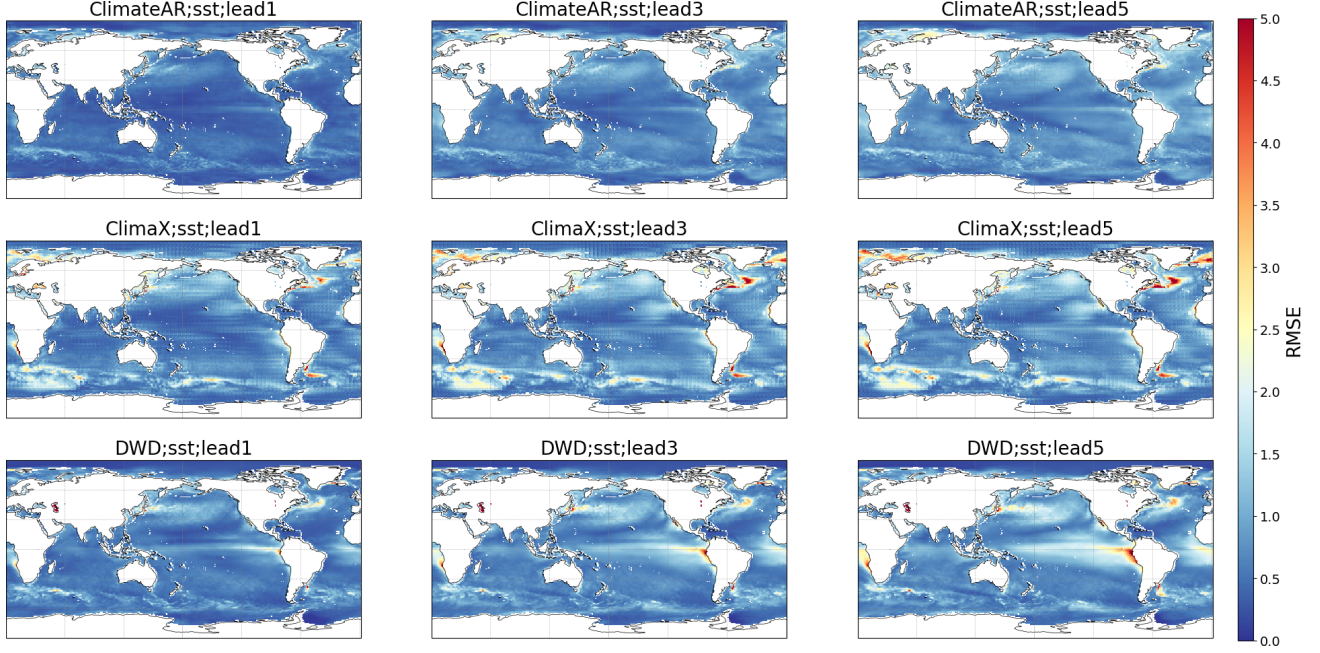


Figure 12. The global RMSE distribution of sea surface temperature forecasting with 1-, 3-, and 5-month lead times in the testing set.

Table 8. 1 month-ahead prediction ACC for different hyperparameters of ClimateAR.

	ACC ($\uparrow$)											
	noise				codebooks				codebook size			
	0.2	0.3	0.4	0.5	2	4	8	16	1024	2048	4096	8192
z500	0.408	0.428	0.414	0.405	0.395	0.394	0.428	0.398	0.402	0.408	0.428	0.397
t2m	0.456	0.480	0.462	0.466	0.445	0.455	0.480	0.467	0.463	0.467	0.480	0.466
pr	0.162	0.190	0.171	0.164	0.159	0.162	0.190	0.164	0.164	0.162	0.190	0.161
psl	0.348	0.374	0.356	0.344	0.327	0.331	0.374	0.335	0.350	0.350	0.374	0.330

Table 9. Result of efficiency study of ClimateAR and data-driven baselines.

Method	Parameters	GPU Memory (GB)	FLOPs	Training Time Min / Epoch	Inference Time s / Step	ACC
ClimaX	104.0 M	19.27	420.97 G	18.78	0.162	0.376
Oneforecast	24.7 M	38.95	548.78 G	69.73	0.516	0.386
GraphCast	28.9 M	34.59	1852.26 G	32.33	0.279	0.376
Pangu	23.9 M	7.60	213.82 G	5.48	0.047	0.329
ClimateAR	480.6 M	18.34	324.23 G	13.13	0.179	0.464

number of predicted members in the set is M , the complexities increases to $O(Mh^2w^2)$.

From Table 9 we can find that: (1) as a generative model, ClimateAR has a lower cost than most deterministic models, which demonstrates a competitive efficiency; (2) due to the patch-based processing, the training time of ClimaX, Pangu, and ClimateAR is significantly lower than that of deterministic graph-based models.

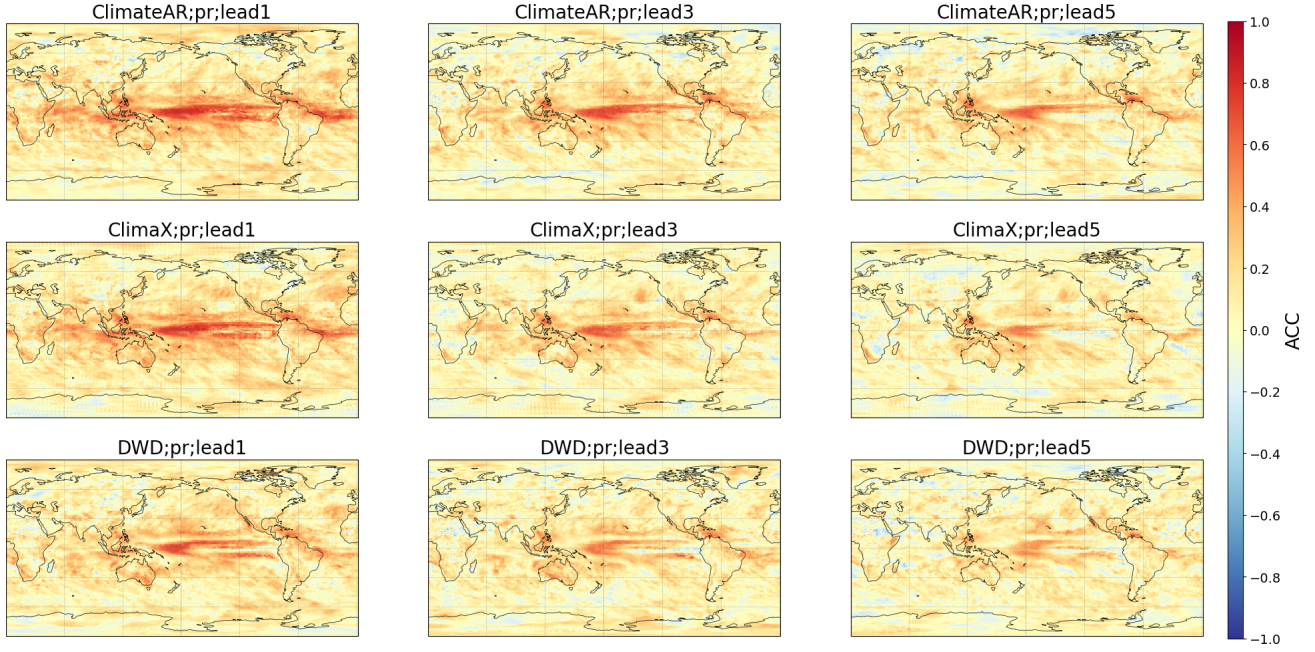


Figure 13. The global ACC distribution of precipitation rate forecasting with 1-, 3-, and 5-month lead times in the testing set.

A.5. Additional Related Work

AutoRegressive Models. AutoRegressive models have achieved significant success in language and visual generation tasks (Radford et al., 2019; Bai et al., 2024). Early visual AR models generate images at pixel level row-by-row (Van den Oord et al., 2016; Van Den Oord et al., 2016; Chen et al., 2020), whereas later models inspired by ViT (Dosovitskiy et al., 2021) generate images at patch level row-by-row (Esser et al., 2021; Lee et al., 2022). Recent advances in visual AR models based on next-scale prediction (Tian et al., 2024) have significantly improved generative quality and speed, and demonstrated scaling capabilities comparable to large language models (LLMs) (Kaplan et al., 2020; Hoffmann et al., 2022). Compared to diffusion models, AR models exhibit superior efficiency in both training and inference, and are particularly advantageous for fusing complex multimodal conditional information with discrete tokens (Yu et al., 2023; Lu et al., 2023). While diffusion-based generative models (Price et al., 2025; Oskarsson et al., 2024) and deterministic models (Nguyen et al., 2023; Bi et al., 2023) have been extensively explored for weather and climate forecasting, the potential of AR generative models in long-term climate forecasting remains underexplored.

A.6. Limitation and Future Work

Despite such advantages, there are several directions that remain to be explored in the future: (1) fine-tuning the model to reduce the cumulative loss in iterative forecasts and extend the forecasting horizon; (2) incorporating contrastive learning or classifier-free guidance techniques in generative models to enhance the model’s ability to predict anomalies; and (3) using ClimateAR to assist in more tasks than forecasting such as dimensionality reduction analysis and anomaly detection.

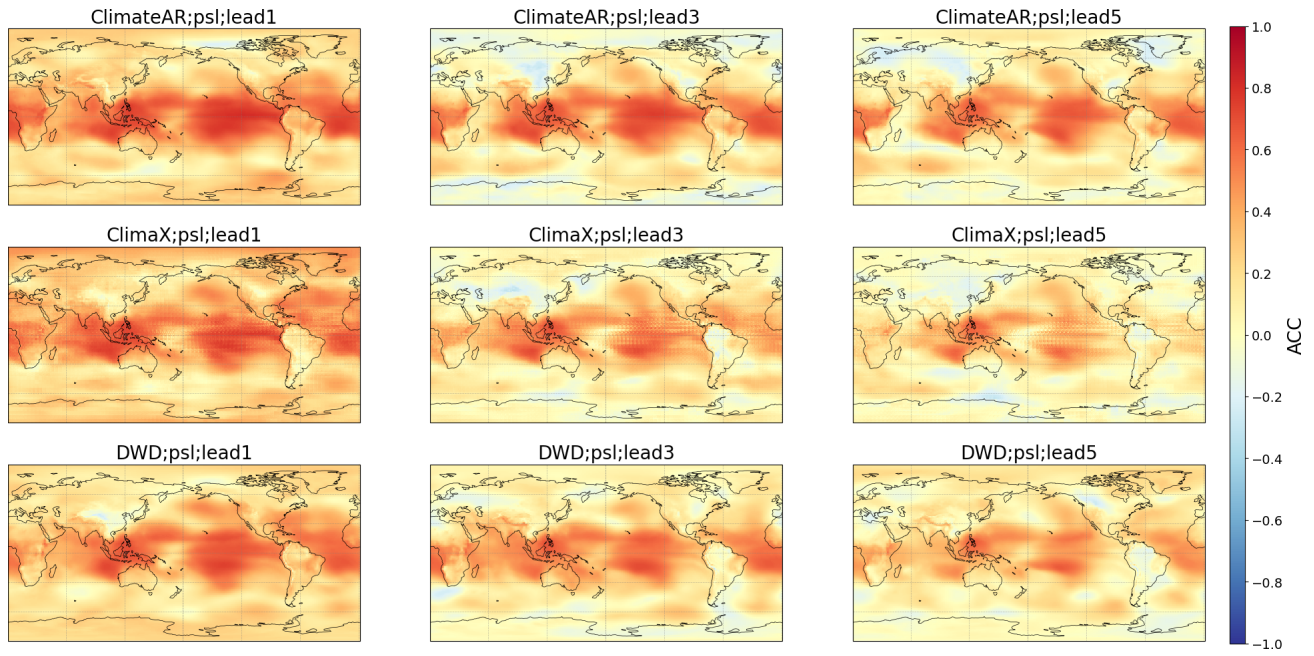


Figure 14. The global ACC distribution of sea level pressure forecasting with 1-, 3-, and 5-month lead times in the testing set.

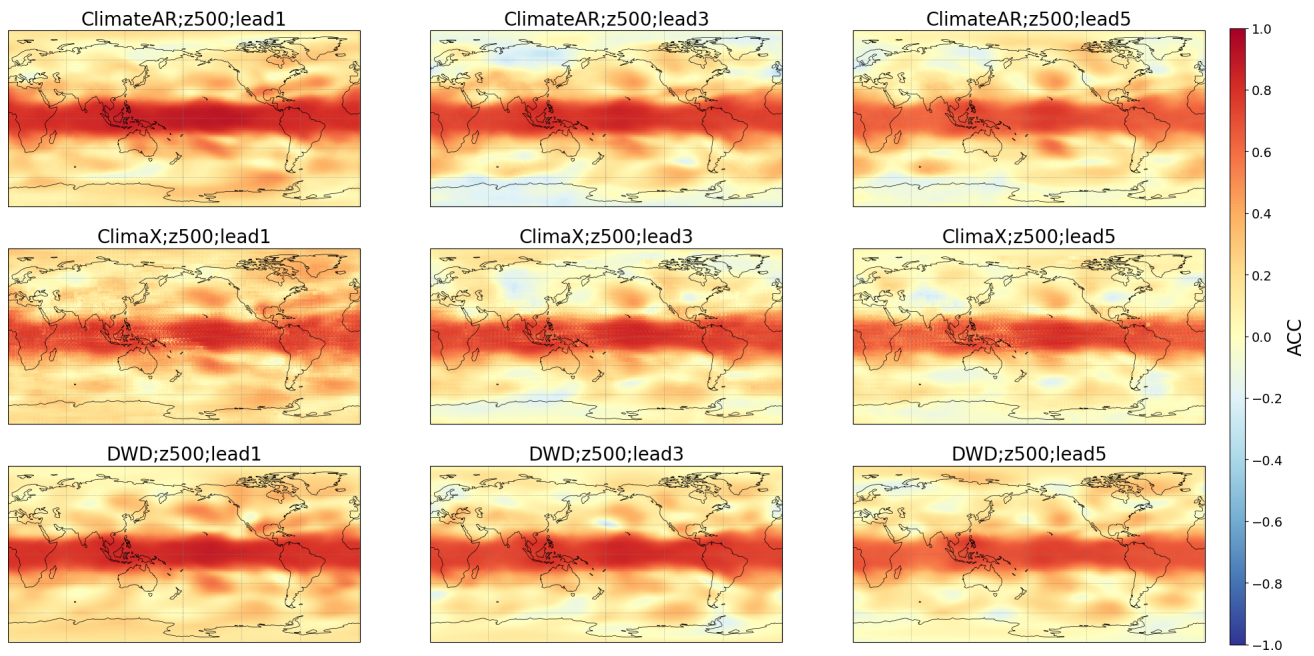


Figure 15. The global ACC distribution of 500hPa geopotential forecasting with 1-, 3-, and 5-month lead times in the testing set.

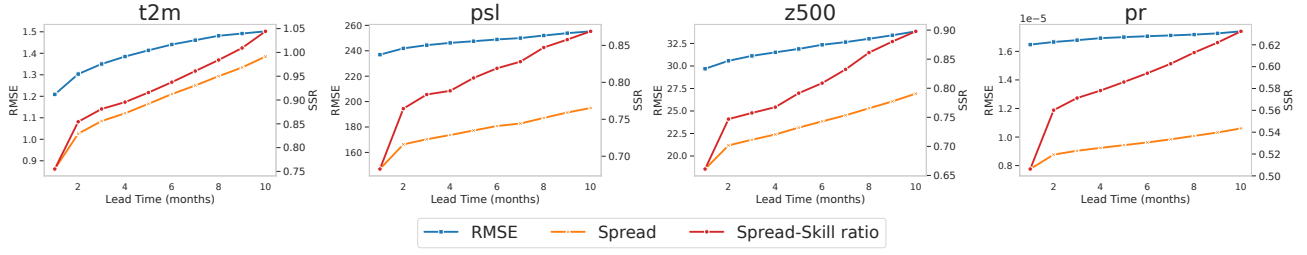


Figure 16. The evolution of the ensemble forecast spread, RMSE, and the spread-skill ratio with lead time.

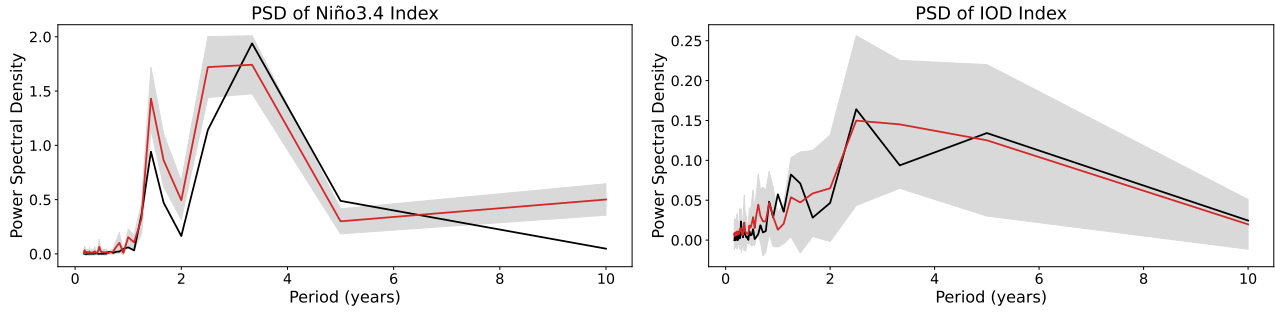


Figure 17. The Power Spectral Density (PSD) of the Niño3.4 index generated by ClimateAR at 6-month lead time and IOD index generated by ClimateAR at 1-month lead time and compare it with the reference data. The shaded area represents the range indicating twice the ensemble standard deviation.

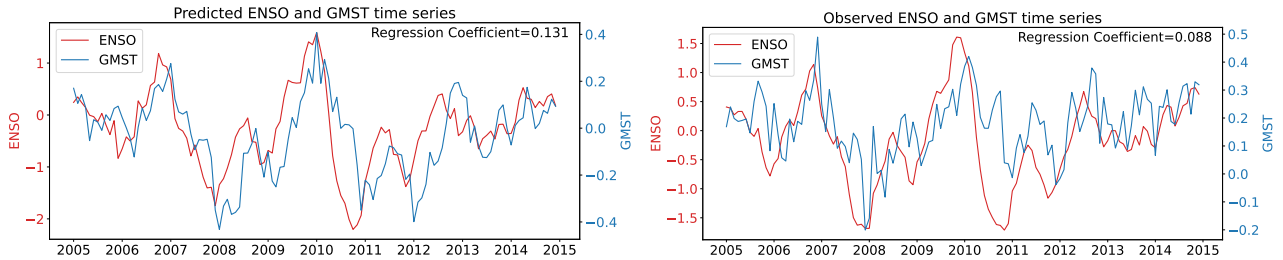


Figure 18. Comparison of the prediction ahead 1 month and observation for the global mean surface temperature (GMST) and ENSO time series.

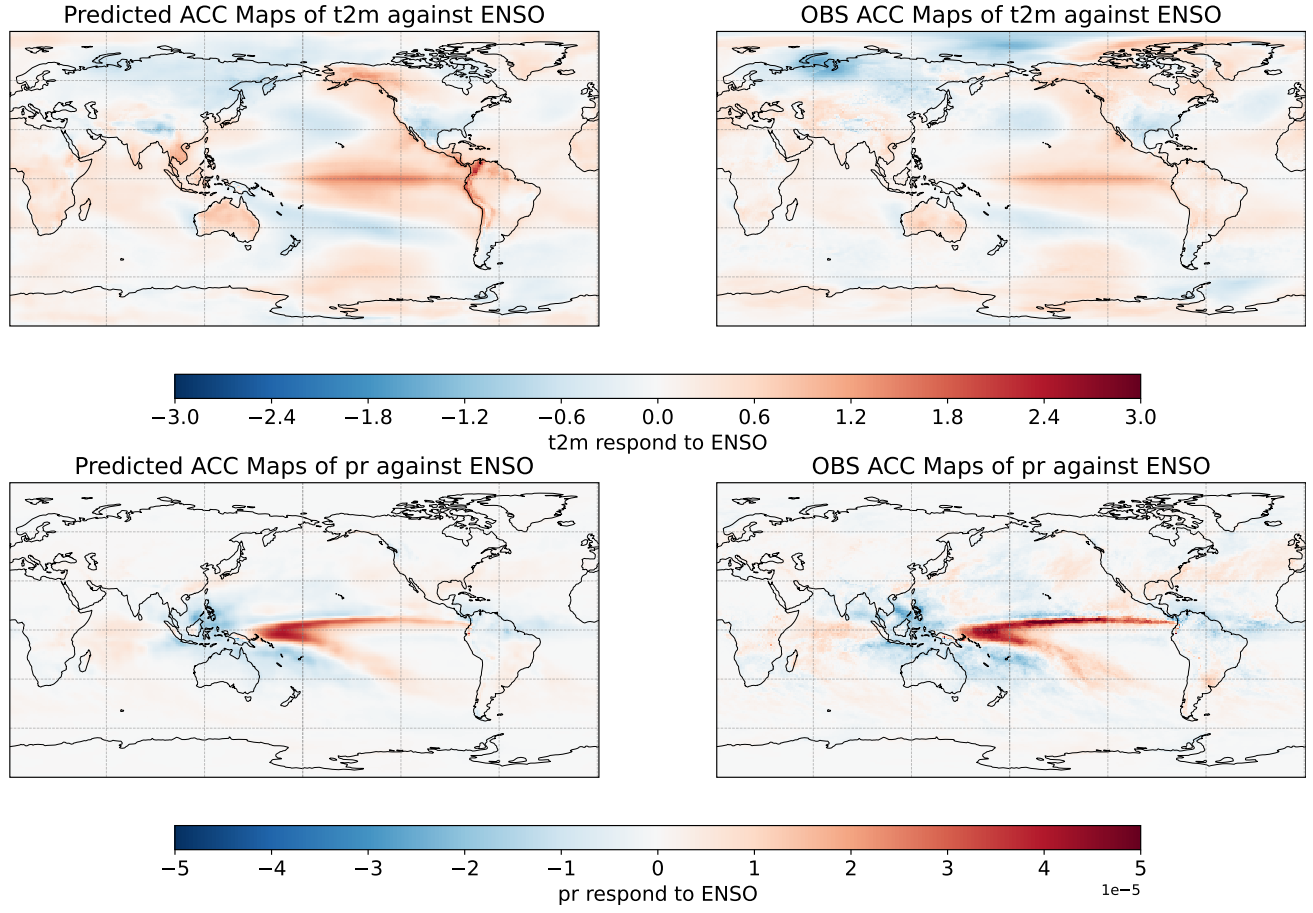


Figure 19. Maps of regression coefficients of predicted ahead 1 month and observed pr and t2m response to the Niño3.4 index over 2005–2014.

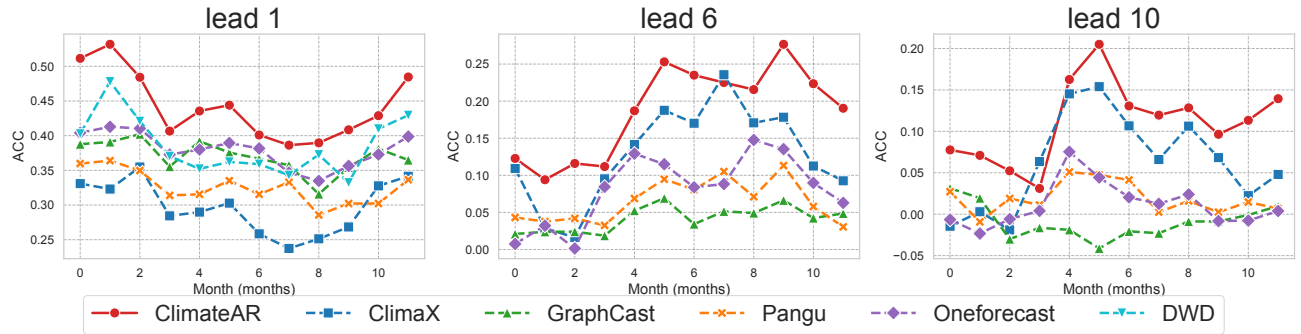


Figure 20. T2m ACC forecast starting from each month at 1-, 6-, and 10-lead times.

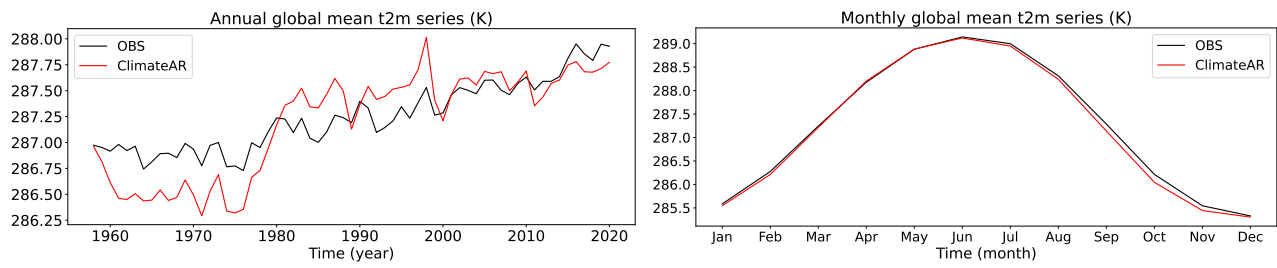


Figure 21. The Global annual-mean and monthly-mean t2m time series of ClimateAR and observation over 1958–2020.